
Discrete Mathematics

Logic • Discrete Structures • Counting • Proofs and Recursion

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Part I

Foundations of Logic

Introduction to Discrete Mathematics

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1.1 What is discrete mathematics?

Discrete mathematics is the study of mathematical structures that are fundamentally *discrete* rather than continuous: integers, finite sets, graphs, logical statements, sequences and algorithms. Where calculus deals with quantities that vary smoothly, discrete mathematics deals with objects that can be counted, listed, and manipulated one at a time.

Definition 1.1. *Discrete mathematics* is the branch of mathematics dealing with countable structures — objects that can assume only distinct, separated values — as opposed to continuous structures such as the real line.

The natural habitat of discrete mathematics is the world of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ and of finite collections. A digital computer is itself a discrete object: its memory holds finitely many bits, each either 0 or 1, so every question about computation is ultimately a question of discrete mathematics.

1.2 The main questions

Across all its subfields, discrete mathematics keeps returning to a small number of fundamental questions.

- **Existence.** Is there an object with a given property? (Is there a path visiting every vertex of a graph exactly once?)
- **Counting.** How many such objects are there? (How many ways can a committee of 5 be chosen from 30 students?)
- **Optimization.** Which object is best? (Which spanning tree of a network has minimal total cost?)
- **Structure.** How are the objects organised? (How does a partial order arrange the subsets of a set?)

- **Algorithms.** How can the answer be computed, and how fast? (Can satisfiability of a logical formula be decided efficiently?)

1.3 Information, summarization and computation

Three recurring activities motivate the tools we will build.

1.3.1 Information

Discrete structures *encode information*. A truth table encodes the behaviour of a logical circuit; a set encodes membership; a relation encodes links between objects; a binary string encodes a file. Understanding which structures can encode which information — and how compactly — is a central concern.

1.3.2 Summarization

Mathematics compresses infinitely many facts into a single statement. The formula

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

summarises infinitely many arithmetic identities at once. Quantifiers ($\forall$, $\exists$), closed forms and recurrences are all instruments of summarization, and proving such summaries correct is what Part IV of this book is about.

1.3.3 Computation

Every algorithm is a discrete process: a sequence of elementary steps acting on discrete data. Counting the steps (complexity), proving the result correct (invariants, induction) and even defining what “computable” means are tasks for discrete mathematics.

1.4 A brief history

Discrete themes are as old as mathematics itself. Euclid (c. 300 BCE) proved that there are infinitely many primes — a discrete existence theorem we shall reprove in Chapter 12. Combinatorial counting flourished with Pascal and Fermat in the 17th century around games of chance, producing Pascal’s triangle (Chapters 10–11). Euler’s 1736 solution of the Königsberg bridge problem founded graph theory. In the 19th century, Boole turned logic into algebra and Cantor created set theory, giving the subject its modern foundations (Parts I–II). The 20th century brought the decisive turn: Gödel, Turing and Church connected logic to computation, and the arrival of digital computers transformed discrete mathematics from a collection of curiosities into the mathematical backbone of computer science.

1.5 Discrete mathematics today

Modern applications are everywhere:

- **Computer science** — data structures, databases, programming language semantics, verification of software and hardware.
- **Cryptography** — number theory and finite algebra protect every online transaction.

- **Networks** — graph theory models the internet, social networks and transportation systems.
- **Data science** — counting, probability on finite spaces, and combinatorial optimisation underlie machine learning pipelines.
- **Operations research** — scheduling, allocation and routing are discrete optimisation problems.

Continuous mathematics asks “how much?”; discrete mathematics asks “how many?”, “does one exist?”, and “how can we build it?”.

1.6 Why study discrete mathematics?

Beyond its applications, discrete mathematics teaches a craft: the craft of *precise reasoning*. In this course you will learn to

- read and write statements whose meaning is completely unambiguous (Chapters 2–5);
- manipulate the basic structures of all mathematics — sets, relations, functions, sequences (Chapters 6–9);
- count complicated configurations exactly (Chapters 10–11);
- construct proofs that withstand any scrutiny, including proofs by induction and recursive constructions (Chapters 12–14).

These skills transfer directly to programming, to algorithm design and to every mathematical subject you will meet later.

1.7 About this course

The book is organised in four parts.

- **Part I — Foundations of Logic.** Propositions, connectives, truth tables, logical equivalence, predicates, quantifiers and rules of inference: the language in which all later chapters are written.
- **Part II — Discrete Structures.** Sets and their operations, relations, functions, sequences and series: the objects that the language talks about.
- **Part III — Counting.** Permutations, combinations, the binomial theorem and a toolbox of binomial identities.
- **Part IV — Proofs and Recursion.** Proof techniques, mathematical induction in its weak, strong and structural forms, and recursive definitions of functions, sets and structures.

Each chapter ends with a graded set of exercises — tagged **EASY**, **MEDIUM** or **HARD** — followed by complete solutions.

1.8 Who is this course for?

The course assumes no prerequisites beyond secondary-school algebra. It is designed for first-year students of computer science, mathematics, engineering and data science, and for anyone who wants to reason precisely about discrete problems. Mathematical maturity is not required at the start — building it is precisely the point.

Exercises

Exercise 1.1 EASY

Classify each of the following as a *discrete* or a *continuous* question: (a) How many 5-card poker hands are possible? (b) What is the area under the curve $y = x^2$ between 0 and 1? (c) Is there a way to schedule 10 exams in 4 slots with no student conflict? (d) At what speed does a falling object hit the ground?

Exercise 1.2 EASY

Give one example from everyday life of each of the five main questions of discrete mathematics (existence, counting, optimization, structure, algorithms).

Exercise 1.3 MEDIUM

A byte consists of 8 bits, each 0 or 1. How many distinct bytes are there? More generally, how many distinct bit strings of length n are there? Justify your count.

Exercise 1.4 MEDIUM

The formula $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ “summarises infinitely many facts”. Verify it directly for $n = 1, 2, 3, 4$, and explain why no amount of direct verification constitutes a proof for all n .

Exercise 1.5 HARD

Euclid proved there are infinitely many primes. Suppose, for contradiction, that $p_1, p_2, \dots, p_k$ were *all* the primes, and consider $N = p_1 p_2 \cdots p_k + 1$. Show that none of the p_i divides N , and deduce a contradiction. (You may use the fact that every integer greater than 1 has at least one prime divisor.)

Solutions to the Exercises

Solution 1.1. (a) Discrete — a finite count. (b) Continuous — an integral over a real interval. (c) Discrete — existence of a finite arrangement. (d) Continuous — a real-valued physical quantity varying smoothly with time. ■

Solution 1.2. For instance: *existence* — is there a seating plan where no two rivals sit together? *Counting* — how many phone numbers are possible with a given prefix? *Optimization* — which route to work is shortest? *Structure* — how is a family tree organised? *Algorithms* — what procedure sorts a list of names alphabetically? (Any sensible examples are acceptable.) ■

Solution 1.3. Each of the 8 bits can be chosen independently in 2 ways, so there are $2 \cdot 2 \cdots 2 = 2^8 = 256$ bytes. The same argument gives 2^n bit strings of length n : choosing a string means making n independent binary choices, and by the product rule (Chapter 10) the total is 2^n . ■

Solution 1.4. $n = 1$: $1 = \frac{1 \cdot 2}{2}$. $n = 2$: $1 + 2 = 3 = \frac{2 \cdot 3}{2}$. $n = 3$: $1 + 2 + 3 = 6 = \frac{3 \cdot 4}{2}$. $n = 4$: $1 + 2 + 3 + 4 = 10 = \frac{4 \cdot 5}{2}$. Direct verification only checks finitely many instances, while the claim is about *all* natural numbers n ; no finite list of checks can rule out a failure at some larger n . A general argument — such as induction (Chapter 13) or Gauss’s pairing trick — is required. ■

Solution 1.5. Suppose $p_1, \dots, p_k$ are all the primes and let $N = p_1 p_2 \cdots p_k + 1$. For each i , dividing N by p_i leaves remainder 1, since $p_i \mid p_1 \cdots p_k$; hence $p_i \nmid N$. But $N > 1$, so N has some prime divisor q . Then q is a prime different from every p_i , contradicting the assumption that the list contained all primes. Therefore there are infinitely many primes. ■

Propositional Logic

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2.1 Propositions

Definition 2.1 (Proposition). A *proposition* is a declarative sentence that is either true (T) or false (F), but never both and never neither. Propositions are denoted by lowercase letters $p, q, r, \dots$ and take their values in the set $\mathbb{B} = \{T, F\}$ of *truth values*.

Example 2.2. “ $2 + 2 = 4$ ” is a true proposition; “Paris is the capital of Spain” is a false proposition. By contrast, “Close the door” (imperative), “What time is it?” (interrogative) and “ $x + 1 = 0$ ” (free variable — a *predicate*, Chapter 4) are not propositions. The self-referential sentence “This sentence is false” is also excluded: it cannot consistently receive either truth value, violating the principle of bivalence.

Definition 2.3 (Atomic and compound). An *atomic* proposition cannot be decomposed using logical connectives; it is symbolised by a single letter. A *compound* proposition is built from atomic propositions using connectives such as $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$.

Propositional logic studies only how the truth of a compound proposition depends on its atoms; looking *inside* an atom is the job of predicate logic (Chapter 4).

2.1.1 Well-formed formulas

The propositional language can be defined by a grammar: every atomic letter is a well-formed formula (wff); if φ and ψ are wffs, so are $\neg\varphi$, $(\varphi \wedge \psi)$, $(\varphi \vee \psi)$, $(\varphi \rightarrow \psi)$ and $(\varphi \leftrightarrow \psi)$; nothing else is a wff. This inductive definition (a recursive set in the sense of Chapter 14) makes “formula” a precise mathematical object.

Definition 2.4 (Truth assignment). A *truth assignment* (or *valuation*) is a function $v : \{p, q, r, \dots\} \rightarrow \mathbb{B}$ assigning a truth value to every atomic proposition. A valuation extends uniquely to all wffs via the truth tables of the connectives.

2.2 The logical connectives

Definition 2.5 (Negation). The *negation* $\neg p$ (“not p ”) is true exactly when p is false.

Definition 2.6 (Conjunction and disjunction). The *conjunction* $p \wedge q$ (“ p and q ”) is true exactly when both p and q are true. The *disjunction* $p \vee q$ (“ p or q ”) is true exactly when at least one of p, q is true (inclusive or).

Definition 2.7 (Exclusive or). The *exclusive or* $p \oplus q$ is true exactly when p and q have different truth values.

Definition 2.8 (Conditional). The *conditional* (material implication) $p \rightarrow q$ (“if p then q ”) is false only when p is true and q is false; in all other cases it is true. p is the *hypothesis* (antecedent), q the *conclusion* (consequent).

A conditional with a false hypothesis is vacuously true: “if $1 = 2$ then I am the Pope” is a true proposition. The conditional asserts nothing about causality — only that the case (p true, q false) does not occur.

Definition 2.9 (Biconditional). The *biconditional* $p \leftrightarrow q$ (“ p if and only if q ”) is true exactly when p and q have the same truth value.

Definition 2.10 (NAND and NOR). $p \uparrow q \equiv \neg(p \wedge q)$ (NAND, Sheffer stroke) and $p \downarrow q \equiv \neg(p \vee q)$ (NOR, Peirce arrow).

The complete behaviour of the binary connectives:

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \uparrow q$
T	T	T	T	F	T	T	F
T	F	F	T	T	F	F	T
F	T	F	T	T	T	F	T
F	F	F	F	F	T	T	T

2.3 Truth tables

To build the truth table of a compound proposition with n atomic letters, list all 2^n valuations (conventionally from $T \cdots T$ down to $F \cdots F$), then evaluate subformulas from the innermost outward.

Example 2.11. Truth table of $(p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$:

p	q	$p \rightarrow q$	$\neg p$	$\neg p \rightarrow \neg q$	whole formula
T	T	T	F	T	T
T	F	F	F	T	F
F	T	T	T	F	F
F	F	T	T	T	T

The formula is true exactly when p and q agree — it is equivalent to $p \leftrightarrow q$.

2.3.1 Precedence

To limit parentheses, the standard precedence (strongest first) is: $\neg$, then $\wedge$, then $\vee$, then $\rightarrow$, then $\leftrightarrow$. Thus $\neg p \vee q \wedge r \rightarrow s$ reads $((\neg p) \vee (q \wedge r)) \rightarrow s$. The conditional associates to the right: $p \rightarrow q \rightarrow r$ means $p \rightarrow (q \rightarrow r)$. When in doubt, parenthesize.

2.4 Converse, inverse, contrapositive

Definition 2.12. For the conditional $p \rightarrow q$: the *converse* is $q \rightarrow p$; the *inverse* is $\neg p \rightarrow \neg q$; the *contrapositive* is $\neg q \rightarrow \neg p$.

Theorem 2.13. A conditional is logically equivalent to its contrapositive: $p \rightarrow q \equiv \neg q \rightarrow \neg p$. It is *not* in general equivalent to its converse or to its inverse (which are equivalent to each other).

Two classical fallacies arise from confusing these forms: *affirming the consequent* (from $p \rightarrow q$ and q , wrongly concluding p) and *denying the antecedent* (from $p \rightarrow q$ and $\neg p$, wrongly concluding $\neg q$).

2.5 Tautology, contradiction, contingency

Definition 2.14. A compound proposition is a *tautology* if it is true under every valuation; a *contradiction* if it is false under every valuation; a *contingency* if it is neither.

Example 2.15. $p \vee \neg p$ is a tautology (law of excluded middle); $p \wedge \neg p$ is a contradiction; $p \rightarrow q$ is a contingency.

Definition 2.16 (Satisfiability). A formula is *satisfiable* if at least one valuation makes it true. The *SAT problem* asks whether a given formula is satisfiable; it is the canonical NP-complete problem, and a formula with n variables has 2^n candidate valuations — brute force quickly becomes infeasible.

2.6 Logical equivalence

Definition 2.17. Two formulas φ and ψ are *logically equivalent*, written $\varphi \equiv \psi$, if they have the same truth value under every valuation — equivalently, if $\varphi \leftrightarrow \psi$ is a tautology.

Theorem 2.18 (Major equivalence laws). For all formulas p, q, r (with T, F denoting a tautology and a contradiction):

Identity	$p \wedge \mathbf{T} \equiv p, \quad p \vee \mathbf{F} \equiv p$
Domination	$p \vee \mathbf{T} \equiv \mathbf{T}, \quad p \wedge \mathbf{F} \equiv \mathbf{F}$
Idempotence	$p \vee p \equiv p, \quad p \wedge p \equiv p$
Double negation	$\neg(\neg p) \equiv p$
Commutativity	$p \vee q \equiv q \vee p, \quad p \wedge q \equiv q \wedge p$
Associativity	$(p \vee q) \vee r \equiv p \vee (q \vee r)$, and dually for $\wedge$
Distributivity	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$, and dually
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q, \quad \neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption	$p \vee (p \wedge q) \equiv p, \quad p \wedge (p \vee q) \equiv p$
Negation	$p \vee \neg p \equiv \mathbf{T}, \quad p \wedge \neg p \equiv \mathbf{F}$

Theorem 2.19 (Implication and biconditional identities).

$$p \rightarrow q \equiv \neg p \vee q, \quad \neg(p \rightarrow q) \equiv p \wedge \neg q, \quad p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p).$$

Each law is proved by a truth table; once proved, laws may be chained to simplify formulas algebraically, replacing equals by equals.

2.7 Logical consequence

Definition 2.20 (Entailment). A set of formulas Γ *entails* φ , written $\Gamma \models \varphi$, if every valuation making all formulas of Γ true also makes φ true. An *argument* with premises Γ and conclusion φ is *valid* exactly when $\Gamma \models \varphi$.

Validity can always be checked by a truth table, but with many atoms this is exponential; Chapter 5 develops *rules of inference* as a scalable alternative. The most important rules — modus ponens (from p and $p \rightarrow q$, infer q), modus tollens (from $\neg q$ and $p \rightarrow q$, infer $\neg p$), hypothetical syllogism, disjunctive syllogism, addition, simplification, conjunction and resolution — are all entailments and are treated in detail there.

2.8 Normal forms

Definition 2.21. A *literal* is an atom or a negated atom. A formula is in *negation normal form* (NNF) if negations occur only on atoms. It is in *disjunctive normal form* (DNF) if it is a disjunction of conjunctions of literals, and in *conjunctive normal form* (CNF) if it is a conjunction of disjunctions of literals.

Theorem 2.22. Every propositional formula is equivalent to one in CNF and to one in DNF.

Example 2.23 (CNF conversion). Convert $\neg(p \rightarrow q) \vee r$ to CNF:

$$\neg(p \rightarrow q) \vee r \equiv (p \wedge \neg q) \vee r \equiv (p \vee r) \wedge (\neg q \vee r),$$

using Theorem 2.19 and then distributivity. The general algorithm: eliminate $\leftrightarrow$ and $\rightarrow$, push negations inward by De Morgan (reaching NNF), then distribute $\vee$ over $\wedge$.

A DNF can also be read directly off a truth table: each row where the formula is true contributes one conjunction of literals describing that row.

2.9 Functional completeness

Definition 2.24. A set of connectives is *functionally complete* if every Boolean function $\mathbb{B}^n \rightarrow \mathbb{B}$ can be expressed using only connectives from the set.

Theorem 2.25. $\{\neg, \wedge, \vee\}$ is functionally complete (via DNF); so are $\{\neg, \wedge\}$ and $\{\neg, \vee\}$ (by De Morgan); and remarkably each of the singletons $\{\uparrow\}$ and $\{\downarrow\}$ is functionally complete.

For NAND: $\neg p \equiv p \uparrow p$ and $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$, which suffices. This is why entire processors can be built from NAND gates alone.

2.10 Logic gates and circuits

Each connective corresponds to a hardware *gate* (AND, OR, NOT, XOR, NAND, NOR). Compound formulas correspond to combinational circuits. The *half-adder*, which adds two bits, illustrates the correspondence: the sum bit is $s = p \oplus q$ and the carry bit is $c = p \wedge q$. Cascading half-adders yields full adders and ultimately arithmetic units — arithmetic reduced to logic.

2.11 Translating English to logic

Useful patterns, with p “it rains” and q “I take an umbrella”:

- “If it rains, I take an umbrella”: $p \rightarrow q$.
- “I take an umbrella only if it rains”: $q \rightarrow p$.
- “I take an umbrella unless it rains”: $\neg p \rightarrow q$.
- “Taking an umbrella is necessary for rain”: $p \rightarrow q$ (q is the necessary condition).
- “Taking an umbrella is sufficient for rain”: $q \rightarrow p$.
- “I take an umbrella if and only if it rains”: $q \leftrightarrow p$.

“Only if” introduces the conclusion, not the hypothesis; “unless” means “if not”. These two patterns cause most translation errors.

2.12 Summary

Propositions take values in $\mathbb{B} = \{\mathbf{T}, \mathbf{F}\}$; connectives $\neg, \wedge, \vee, \oplus, \rightarrow, \leftrightarrow, \uparrow, \downarrow$ combine them. Truth tables decide every semantic question in principle; equivalence laws (Theorem 2.18) decide them algebraically. Every formula has a CNF and a DNF; $\{\uparrow\}$ alone suffices to express everything; and logic realised in silicon becomes digital circuitry.

Exercises

Exercise 2.1 EASY

Which of the following are propositions? (a) “7 is prime.” (b) “ $x^2 \geq 0$.” (c) “Read this chapter twice.” (d) “ $1 + 1 = 3$.” (e) “This statement is false.”

Exercise 2.2 EASY

Let p : “the file compiles” and q : “the tests pass”. Express in symbols: (a) the file compiles but the tests fail; (b) if the file does not compile then the tests fail; (c) the tests pass only if the file compiles; (d) the file compiles if and only if the tests pass.

Exercise 2.3 EASY

Build the truth table of $(p \vee q) \wedge \neg(p \wedge q)$ and identify which standard connective it represents.

Exercise 2.4 MEDIUM

State the converse, inverse and contrapositive of: “If n is divisible by 6, then n is divisible by 3.” Which of the four statements are true for all integers n ?

Exercise 2.5 MEDIUM

Classify each formula as tautology, contradiction or contingency: (a) $(p \rightarrow q) \vee (q \rightarrow p)$; (b) $(p \wedge \neg p) \rightarrow q$; (c) $(p \rightarrow q) \rightarrow p$.

Exercise 2.6 MEDIUM

Using the equivalence laws (not truth tables), simplify $\neg(p \vee \neg q) \vee (\neg p \wedge \neg q)$ as far as possible, citing each law used.

Exercise 2.7 MEDIUM

Prove using equivalence laws that $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$.

Exercise 2.8 MEDIUM

Convert $(p \wedge q) \rightarrow r$ to CNF, then to DNF, showing each step.

Exercise 2.9 HARD

Show that $\{\rightarrow, F\}$ is functionally complete, where F is the constant false. *Hint: first express $\neg p$, then $\vee$, then $\wedge$.*

Exercise 2.10 HARD

Express $p \vee q$ and $p \rightarrow q$ using only the NOR connective $\downarrow$.

Exercise 2.11 HARD

A formula has the truth table that is true exactly on the rows $(p, q, r) = (T, T, F)$ and (F, T, T) . Write its DNF, and give an equivalent CNF.

Exercise 2.12 HARD

Three suspects make statements. Alice: “Bob is lying.” Bob: “Carol is lying.” Carol: “Alice and Bob are both lying.” Using propositional variables a, b, c for “Alice (resp. Bob, Carol) tells the truth”, formalise the situation and determine who is telling the truth.

Solutions to the Exercises

Solution 2.1. (a) Yes (true). (b) No as stated — x is a free variable, so it is a predicate; (it becomes a proposition once quantified, e.g. $\forall x \in \mathbb{R}, x^2 \geq 0$). (c) No — imperative. (d) Yes (false). (e) No — it can receive no consistent truth value. ■

Solution 2.2. (a) $p \wedge \neg q$. (b) $\neg p \rightarrow \neg q$. (c) $q \rightarrow p$. (d) $p \leftrightarrow q$. ■

Solution 2.3.

p	q	$p \vee q$	$\neg(p \wedge q)$	formula
T	T	T	F	F
T	F	T	T	T
F	T	T	T	T
F	F	F	T	F

True exactly when p and q differ: this is the exclusive or $p \oplus q$. ■

Solution 2.4. Converse: “if n is divisible by 3 then n is divisible by 6” — false ($n = 9$). Inverse: “if n is not divisible by 6 then n is not divisible by 3” — false ($n = 9$ again). Contrapositive: “if n is not divisible by 3 then n is not divisible by 6” — true, being equivalent to the original true statement. ■

Solution 2.5. (a) Tautology: if $p \rightarrow q$ fails then p is true and q false, making $q \rightarrow p$ true. (b) Tautology: the hypothesis $p \wedge \neg p$ is always false, so the conditional is vacuously true. (c) Contingency: true when $p = \text{T}$; false when $p = \text{F}$ (then $p \rightarrow q$ is true but the conclusion p is false). ■

Solution 2.6.

$$\begin{aligned}
 \neg(p \vee \neg q) \vee (\neg p \wedge \neg q) &\equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q) && \text{De Morgan, double negation} \\
 &\equiv \neg p \wedge (q \vee \neg q) && \text{distributivity} \\
 &\equiv \neg p \wedge \text{T} && \text{negation law} \\
 &\equiv \neg p && \text{identity law.}
 \end{aligned}$$

Solution 2.7.

$$\begin{aligned}
 (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{implication identity} \\
 &\equiv (\neg p \wedge \neg q) \vee r && \text{distributivity} \\
 &\equiv \neg(p \vee q) \vee r && \text{De Morgan} \\
 &\equiv (p \vee q) \rightarrow r && \text{implication identity.}
 \end{aligned}$$

Solution 2.8. $(p \wedge q) \rightarrow r \equiv \neg(p \wedge q) \vee r \equiv \neg p \vee \neg q \vee r$. This single clause is simultaneously a CNF (one disjunctive clause) and a DNF (disjunction of three one-literal conjunctions). ■

Solution 2.9. $\neg p \equiv p \rightarrow \mathbf{F}$. Then $p \vee q \equiv \neg p \rightarrow q \equiv (p \rightarrow \mathbf{F}) \rightarrow q$, and $p \wedge q \equiv \neg(\neg p \vee \neg q) \equiv (((p \rightarrow \mathbf{F}) \rightarrow (q \rightarrow \mathbf{F}))) \rightarrow \mathbf{F}$ after expanding $\neg p \vee \neg q \equiv p \rightarrow \neg q$, i.e. $p \wedge q \equiv (p \rightarrow (q \rightarrow \mathbf{F})) \rightarrow \mathbf{F}$. Since $\{\neg, \wedge, \vee\}$ is functionally complete, so is $\{\rightarrow, \mathbf{F}\}$. ■

Solution 2.10. $\neg x \equiv x \downarrow x$. Then $p \vee q \equiv \neg(p \downarrow q) \equiv (p \downarrow q) \downarrow (p \downarrow q)$, and $p \rightarrow q \equiv \neg p \vee q$, so with $A = p \downarrow p (= \neg p)$: $p \rightarrow q \equiv (A \downarrow q) \downarrow (A \downarrow q)$. ■

Solution 2.11. DNF: each true row becomes a conjunction: $(p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r)$. For a CNF, note both rows have $q = \mathbf{T}$, and $p \leftrightarrow \neg r$ holds on them (p and r differ); conversely $q \wedge (p \oplus r)$ is true exactly on these two rows. Hence a CNF is $q \wedge (p \vee r) \wedge (\neg p \vee \neg r)$. ■

Solution 2.12. The statements give $a \leftrightarrow \neg b$, $b \leftrightarrow \neg c$, $c \leftrightarrow (\neg a \wedge \neg b)$. From the first two: a and b differ, b and c differ, so $a = c$ and $a \neq b$. Case $a = \mathbf{T}$: then $c = \mathbf{T}$, but Carol's statement requires $\neg a \wedge \neg b$, i.e. $a = \mathbf{F}$ — contradiction. Case $a = \mathbf{F}$: then $b = \mathbf{T}$, $c = \mathbf{F}$. Check Carol: she lies, and indeed $\neg a \wedge \neg b = \mathbf{F}$ — consistent. So Bob alone tells the truth. ■

Equivalence of Propositions

Notes By Pr. El Hadiq Zouhair

3.1 Overview

Chapter 2 introduced logical equivalence through truth tables. This chapter turns equivalence into a working calculus: classifying formulas by their satisfiability profile, simplifying them with the standard laws, and putting them into *normal forms* suited to mechanical processing.

3.2 Tautology, contradiction, contingency revisited

Definition 3.1. Let φ be a formula over n atomic letters, with 2^n valuations. φ is a *tautology* if all 2^n valuations satisfy it, a *contradiction* if none does, and a *contingency* otherwise. φ is *satisfiable* when at least one valuation satisfies it — i.e. exactly when it is not a contradiction.

Example 3.2. $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ is a tautology (the contrapositive law); $(p \oplus q) \wedge (p \leftrightarrow q)$ is a contradiction; $p \rightarrow (q \wedge r)$ is a contingency.

φ is a tautology if and only if $\neg\varphi$ is a contradiction. Tautology checking and satisfiability checking are therefore two faces of the same computational problem.

3.3 The main equivalences

The equivalence laws of Chapter 2 — identity, domination, idempotence, double negation, commutativity, associativity, distributivity, De Morgan, absorption, negation — together with the implication identities

$$p \rightarrow q \equiv \neg p \vee q, \quad \neg(p \rightarrow q) \equiv p \wedge \neg q, \quad p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p),$$

form a complete working kit. Two further conditional equivalences are worth memorising:

$$p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r \quad (\text{exportation}),$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r), \quad (p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r.$$

Example 3.3 (Simplifying a proposition). Simplify $\varphi = (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$.

$$\begin{aligned} \varphi &\equiv (p \wedge (q \vee \neg q)) \vee (\neg p \wedge q) && \text{distributivity} \\ &\equiv p \vee (\neg p \wedge q) && \text{negation, identity} \\ &\equiv (p \vee \neg p) \wedge (p \vee q) && \text{distributivity} \\ &\equiv p \vee q && \text{negation, identity.} \end{aligned}$$

Three two-literal clauses collapse to a single disjunction.

Such simplification is exactly what automated tools (SAT preprocessors, circuit minimisers, symbolic algebra systems) perform on a large scale; the Quine–McCluskey algorithm and Karnaugh maps systematise it for circuit design.

3.4 Disjunctive normal form

Definition 3.4. A *minterm* over atoms $p_1, \dots, p_n$ is a conjunction containing each p_i exactly once, either plain or negated. A formula is in *disjunctive normal form* (DNF) if it is a disjunction of conjunctions of literals; it is in *full* (canonical) DNF if every disjunct is a minterm.

Theorem 3.5 (DNF from the truth table). Every satisfiable formula φ is equivalent to the disjunction of the minterms of the rows where φ is true. Hence every formula has a full DNF, unique up to order.

Example 3.6 (XOR: “for me or against me”). The exclusive or is true on rows (T, F) and (F, T), so its full DNF is

$$p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q).$$

Example 3.7 (A counting function). Define $m(p, q, r)$ to be true when *at least two* of p, q, r are true (the “majority” function). The true rows are (T, T, T), (T, T, F), (T, F, T), (F, T, T), giving the full DNF

$$m \equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r),$$

which simplifies (by absorption-style grouping) to $(p \wedge q) \vee (p \wedge r) \vee (q \wedge r)$.

3.5 Other normal forms

Definition 3.8. A formula is in *negation normal form* (NNF) if negation appears only on atoms, and in *conjunctive normal form* (CNF) if it is a conjunction of *clauses*, each clause a disjunction of literals. The full CNF uses *maxterms* — clauses containing every atom once — one for each *false* row of the truth table.

CNF is the input format of SAT solvers and of the resolution proof system; DNF makes satisfiability trivial to read off but may be exponentially large.

3.5.1 CNF $\leftrightarrow$ DNF conversion

The two directions are symmetric: distribute $\vee$ over $\wedge$ to go from CNF to DNF, and $\wedge$ over $\vee$ for the converse.

Example 3.9.

$$(p \vee q) \wedge (\neg p \vee r) \equiv (p \wedge \neg p) \vee (p \wedge r) \vee (q \wedge \neg p) \vee (q \wedge r) \equiv (p \wedge r) \vee (\neg p \wedge q) \vee (q \wedge r),$$

a CNF of two clauses becoming a DNF of three conjunctions (the contradictory term vanishes). The term $q \wedge r$ is in fact absorbed by no other and must stay; in general the blow-up can be exponential.

3.6 Satisfiability research

Deciding satisfiability of a CNF formula is the historic NP-complete problem (Cook–Levin, 1971). Despite the exponential worst case, modern conflict-driven clause-learning (CDCL) solvers routinely handle industrial instances with millions of variables, powering hardware verification, planning and dependency resolution. The special case 2-SAT (clauses of width 2) is solvable in linear time, while 3-SAT is already NP-complete — a remarkably sharp frontier.

3.7 Summary

Formulas split into tautologies, contradictions and contingencies; the equivalence laws constitute an algebra for simplifying them; every formula has a CNF and a DNF, readable from the false and true rows of its truth table respectively; and satisfiability, trivial for DNF, is the canonical hard problem for CNF.

Exercises

Exercise 3.1 EASY

Classify as tautology, contradiction or contingency: (a) $(p \wedge q) \rightarrow p$; (b) $p \leftrightarrow \neg p$; (c) $(p \vee q) \wedge \neg p$.

Exercise 3.2 EASY

Verify by truth table the exportation law $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$.

Exercise 3.3 MEDIUM

Simplify using the laws, citing each step: $\neg(\neg p \wedge \neg q) \wedge (p \vee \neg q)$.

Exercise 3.4 MEDIUM

Simplify $(p \rightarrow q) \wedge (p \rightarrow \neg q)$ to a formula with a single connective.

Exercise 3.5 MEDIUM

Write the full DNF of $p \rightarrow \neg q$ (two atoms), directly from its truth table.

Exercise 3.6 MEDIUM

Give the full CNF of $p \oplus q$, using the false rows of its truth table.

Exercise 3.7 HARD

Let $f(p, q, r)$ be true exactly when an *odd* number of p, q, r are true. Write its full DNF and show $f \equiv p \oplus q \oplus r$.

Exercise 3.8 HARD

Convert $(p \wedge q) \vee (r \wedge s)$ into CNF, and explain why a CNF of a DNF with k conjunctions of two literals each may need 2^k clauses.

Exercise 3.9 HARD

Show that the majority function $m \equiv (p \wedge q) \vee (p \wedge r) \vee (q \wedge r)$ is also equivalent to the CNF $(p \vee q) \wedge (p \vee r) \wedge (q \vee r)$.

Solutions to the Exercises

Solution 3.1. (a) Tautology — if $p \wedge q$ holds, p holds. (b) Contradiction — the two sides always differ. (c) Contingency — true for $p = \text{F}, q = \text{T}$, false for $p = \text{T}$. ■

Solution 3.2. Both sides are false exactly on the row $(p, q, r) = (\text{T}, \text{T}, \text{F})$: the left side because $q \rightarrow r$ then fails with p true; the right side because $p \wedge q$ is true and r false. All other rows make both sides true, hence equivalence. ■

Solution 3.3.

$$\begin{aligned}
 \neg(\neg p \wedge \neg q) \wedge (p \vee \neg q) &\equiv (p \vee q) \wedge (p \vee \neg q) && \text{De Morgan, double negation} \\
 &\equiv p \vee (q \wedge \neg q) && \text{distributivity} \\
 &\equiv p \vee \text{F} \equiv p && \text{negation, identity.}
 \end{aligned}$$

■

Solution 3.4. $(p \rightarrow q) \wedge (p \rightarrow \neg q) \equiv p \rightarrow (q \wedge \neg q) \equiv p \rightarrow \text{F} \equiv \neg p$. ■

Solution 3.5. $p \rightarrow \neg q$ is true on rows $(\text{T}, \text{F}), (\text{F}, \text{T}), (\text{F}, \text{F})$, so

$$p \rightarrow \neg q \equiv (p \wedge \neg q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q).$$

■

Solution 3.6. $p \oplus q$ is false on rows (T, T) and (F, F) . Each false row yields the maxterm that negates it: $(\neg p \vee \neg q)$ for (T, T) and $(p \vee q)$ for (F, F) . Hence $p \oplus q \equiv (p \vee q) \wedge (\neg p \vee \neg q)$. ■

Solution 3.7. Odd-parity rows: exactly one true — $(\text{T}, \text{F}, \text{F}), (\text{F}, \text{T}, \text{F}), (\text{F}, \text{F}, \text{T})$ — or all three true $(\text{T}, \text{T}, \text{T})$. Full DNF:

$$(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge r).$$

Since $\oplus$ is associative and $x \oplus y$ is true when exactly one of x, y is true, $(p \oplus q) \oplus r$ is true when the number of true atoms among p, q, r is odd — the same four rows. Hence $f \equiv p \oplus q \oplus r$. ■

Solution 3.8. Distribute:

$$(p \wedge q) \vee (r \wedge s) \equiv (p \vee r) \wedge (p \vee s) \wedge (q \vee r) \wedge (q \vee s).$$

With k two-literal conjunctions $(a_1 \wedge b_1) \vee \cdots \vee (a_k \wedge b_k)$, distributing chooses one literal from each conjunct in every clause: 2^k clauses, and in general (for distinct variables) none is redundant. ■

Solution 3.9. Expand the CNF: $(p \vee q) \wedge (p \vee r) \wedge (q \vee r)$. If at least two of p, q, r are true, each clause contains a true literal, so the CNF is true. If at most one is true, some pair among $\{p, q\}, \{p, r\}, \{q, r\}$ consists of two false atoms, and the corresponding clause fails. So the CNF is true exactly when at least two atoms are true — the majority function m . ■

Predicates and Quantifiers

Notes By Pr. El Hadiq Zouhair

4.1 Overview

Propositional logic cannot look inside a statement: from “all men are mortal” and “Socrates is a man” it cannot extract “Socrates is mortal”. Predicate logic repairs this by introducing *variables*, *predicates* and *quantifiers*, turning logic into a language rich enough for all of mathematics.

4.2 Predicates

Definition 4.1 (Predicate). A *predicate* is a statement $P(x_1, \dots, x_n)$ containing variables, which becomes a proposition once each variable is given a value from its *domain* (universe of discourse) U . A one-variable predicate is called a *propositional function* $P : U \rightarrow \mathbb{B}$.

Example 4.2. Over $U = \mathbb{Z}$, let $P(x)$: “ $x^2 \leq 9$ ”. Then $P(2)$ is true, $P(5)$ is false; $P(x)$ alone is neither — it is a function awaiting an argument.

Example 4.3 (Pythagorean triples). Let $T(a, b, c)$: “ $a^2 + b^2 = c^2$ ” over positive integers. Then $T(3, 4, 5)$ and $T(5, 12, 13)$ are true while $T(2, 3, 4)$ is false. The predicate packages infinitely many propositions into a single formula.

There are two ways to convert a predicate into a proposition: substitute concrete values, or *quantify* the variables.

4.3 The universal quantifier

Definition 4.4. The *universal quantification* $\forall x P(x)$ (“for all x , $P(x)$ ”) is true exactly when $P(a)$ is true for *every* element a of the domain. An element a with $P(a)$ false is a *counterexample* and makes $\forall x P(x)$ false.

Example 4.5. Over $U = \mathbb{R}$: $\forall x (x^2 \geq 0)$ is true. Over $U = \mathbb{Z}$: $\forall x (x^2 > 0)$ is false — the single counterexample $x = 0$ suffices.

When the domain is finite, $U = \{a_1, \dots, a_n\}$, the universal quantifier is a big conjunction: $\forall x P(x) \equiv P(a_1) \wedge \dots \wedge P(a_n)$.

4.4 The existential quantifier

Definition 4.6. The *existential quantification* $\exists x P(x)$ (“there exists an x such that $P(x)$ ”) is true exactly when $P(a)$ is true for *at least one* element a of the domain; a single *witness* suffices.

Example 4.7. Over $U = \mathbb{Z}$: $\exists x (x^2 = 4)$ is true (witness $x = 2$, or -2); $\exists x (x^2 = 3)$ is false.

Over a finite domain the existential quantifier is a big disjunction: $\exists x P(x) \equiv P(a_1) \vee \dots \vee P(a_n)$. The uniqueness variant $\exists! x P(x)$ asserts that exactly one witness exists.

The truth of a quantified statement depends on the domain. “ $\exists x (2x = 1)$ ” is false over $\mathbb{Z}$ but true over $\mathbb{Q}$. Always state the universe.

4.5 Quantifier equivalences

Theorem 4.8 (De Morgan laws for quantifiers).

$$\neg \forall x P(x) \equiv \exists x \neg P(x), \quad \neg \exists x P(x) \equiv \forall x \neg P(x).$$

Negation flips each quantifier as it moves inward. To negate a quantified statement: change every $\forall$ to $\exists$ and vice versa, then negate the matrix.

Theorem 4.9 (Distribution laws). $\forall$ distributes over $\wedge$, and $\exists$ over $\vee$:

$$\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x), \quad \exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x).$$

Neither distributes over the other connective: $\forall x (P(x) \vee Q(x))$ is *not* equivalent to $\forall x P(x) \vee \forall x Q(x)$, and dually.

Example 4.10. Over $\mathbb{Z}$, with $P(x)$: “ x is even” and $Q(x)$: “ x is odd”: $\forall x (P(x) \vee Q(x))$ is true, but $\forall x P(x) \vee \forall x Q(x)$ is false.

Restricted quantification abbreviates implication and conjunction:

$$\forall x \in S, P(x) \text{ means } \forall x (x \in S \rightarrow P(x)), \quad \exists x \in S, P(x) \text{ means } \exists x (x \in S \wedge P(x)).$$

4.6 Nested quantifiers

Quantifiers may be stacked; **order matters** when the quantifiers differ.

Example 4.11. Over $\mathbb{R}$:

$$\forall x \exists y (x + y = 0)$$

is true ($y = -x$ depends on x), while

$$\exists y \forall x (x + y = 0)$$

is false (no single y works for all x). Same-type quantifiers commute: $\forall x \forall y \equiv \forall y \forall x$.

Example 4.12 (The limit definition). “The sequence (a_n) converges to L ” is the triply-quantified statement

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, \quad |a_n - L| < \varepsilon,$$

whose negation, by repeated use of Theorem 4.8, is $\exists \varepsilon > 0 \forall N \exists n \geq N, |a_n - L| \geq \varepsilon$. Fluency with nested quantifiers is exactly what analysis requires (Chapter 9).

4.7 Translating statements

Example 4.13. Domain: all people. $S(x)$: “ x is a student”; $C(x)$: “ x has a computer”; $F(x, y)$: “ x and y are friends”.

- “Every student has a computer”: $\forall x (S(x) \rightarrow C(x))$.
- “Some student has no computer”: $\exists x (S(x) \wedge \neg C(x))$.
- “Everybody has a friend”: $\forall x \exists y F(x, y)$.
- “There is someone who is friends with everyone”: $\exists x \forall y F(x, y)$.
- “No two distinct people have exactly the same friends”: $\forall x \forall y (x \neq y \rightarrow \exists z (F(x, z) \oplus F(y, z)))$.

Pattern to memorise: universal statements pair with $\rightarrow$ (“all S are C ”: $\forall x(S(x) \rightarrow C(x))$), while existential statements pair with $\wedge$ (“some S is C ”: $\exists x(S(x) \wedge C(x))$). Mixing these up is the most common translation error.

4.8 Summary

Predicates are truth-valued functions of variables; quantifiers $\forall$ and $\exists$ bind those variables to produce propositions. Negation exchanges the two quantifiers (De Morgan); $\forall$ distributes over $\wedge$, $\exists$ over $\vee$; the order of unlike nested quantifiers changes meaning. Universal claims are refuted by one counterexample; existential claims are proved by one witness.

Exercises

Exercise 4.1 EASY

Let $P(x)$: “ $x^2 = x$ ” over the domain $\mathbb{Z}$. Determine the truth values of $P(0)$, $P(1)$, $P(2)$, $\exists x P(x)$ and $\forall x P(x)$.

Exercise 4.2 EASY

Write the negation of each statement, with quantifiers pushed inward: (a) $\forall x (x^2 > x)$; (b) $\exists x (x^2 = 2)$.

Exercise 4.3 EASY

Over the domain of all animals, with $C(x)$: “ x is a cat” and $M(x)$: “ x chases mice”, translate: (a) “every cat chases mice”; (b) “some cat does not chase mice”. Then negate (a) symbolically and in English.

Exercise 4.4 MEDIUM

Determine, with justification, the truth value over $\mathbb{Z}$ of: (a) $\forall n \exists m (m > n)$; (b) $\exists m \forall n (m > n)$; (c) $\forall n \exists m (n + m = 0)$; (d) $\exists m \forall n (nm = n)$.

Exercise 4.5 MEDIUM

Show by a concrete domain and predicates that $\forall x (P(x) \vee Q(x))$ is not equivalent to $\forall x P(x) \vee \forall x Q(x)$, and that $\exists x (P(x) \wedge Q(x))$ is not equivalent to $\exists x P(x) \wedge \exists x Q(x)$.

Exercise 4.6 MEDIUM

Express using quantifiers over $\mathbb{Z}$: (a) “ n is even”; (b) “ n is a perfect square”; (c) “ n is prime” (you may use $|$ for divisibility); (d) “there is no largest integer”.

Exercise 4.7 MEDIUM

Write the negation of the convergence statement $\forall \varepsilon > 0 \exists N \forall n \geq N, |a_n - L| < \varepsilon$, justifying each quantifier flip.

Exercise 4.8 HARD

Let $F(x, y)$: “ x trusts y ” over a set of people. Express: (a) “everyone trusts somebody, but nobody trusts everybody”; (b) “there are two distinct people who trust each other”; (c) “trust is never mutual”. Can (b) and (c) both be true? Explain.

Exercise 4.9 HARD

Prove that $\exists x \forall y P(x, y) \models \forall y \exists x P(x, y)$, and give a counterexample showing the converse entailment fails.

Solutions to the Exercises

Solution 4.1. $x^2 = x$ holds iff $x(x - 1) = 0$ iff $x \in \{0, 1\}$. So $P(0)$, $P(1)$ are true, $P(2)$ is false; $\exists x P(x)$ is true (witness 0); $\forall x P(x)$ is false (counterexample 2). ■

Solution 4.2. (a) $\exists x (x^2 \leq x)$. (b) $\forall x (x^2 \neq 2)$. ■

Solution 4.3. (a) $\forall x (C(x) \rightarrow M(x))$. (b) $\exists x (C(x) \wedge \neg M(x))$. Negation of (a): $\neg \forall x (C(x) \rightarrow M(x)) \equiv \exists x (C(x) \wedge \neg M(x))$ — “some cat does not chase mice”, which is exactly (b). ■

Solution 4.4. (a) True: given n , take $m = n + 1$. (b) False: for any m , $n = m$ fails $m > n$. (c) True: take $m = -n$. (d) True: $m = 1$ satisfies $n \cdot 1 = n$ for all n . ■

Solution 4.5. Over $\mathbb{Z}$ with $P = \text{“even”}$, $Q = \text{“odd”}$: every integer is even or odd, so $\forall x(P \vee Q)$ is true, while neither $\forall xP$ nor $\forall xQ$ holds. For the second pair: $\exists xP(x)$ and $\exists xQ(x)$ are both true, but no integer is both even and odd, so $\exists x(P \wedge Q)$ is false. ■

Solution 4.6. (a) $\exists k(n = 2k)$. (b) $\exists k(n = k^2)$. (c) $n > 1 \wedge \forall d(d \mid n \rightarrow (d = 1 \vee d = n))$. (d) $\neg \exists m \forall n(n \leq m)$, equivalently $\forall m \exists n(n > m)$. ■

Solution 4.7. Negate step by step: $\neg \forall \varepsilon \Phi \equiv \exists \varepsilon \neg \Phi$; $\neg \exists N \Psi \equiv \forall N \neg \Psi$; $\neg \forall n \geq N \chi \equiv \exists n \geq N \neg \chi$; finally $\neg(|a_n - L| < \varepsilon) \equiv |a_n - L| \geq \varepsilon$. Result:

$$\exists \varepsilon > 0 \forall N \exists n \geq N, \quad |a_n - L| \geq \varepsilon.$$

■

Solution 4.8. (a) $\forall x \exists y F(x, y) \wedge \neg \exists x \forall y F(x, y)$. (b) $\exists x \exists y (x \neq y \wedge F(x, y) \wedge F(y, x))$. (c) $\forall x \forall y (x \neq y \rightarrow \neg(F(x, y) \wedge F(y, x)))$. (b) and (c) are direct negations of each other (over pairs of distinct people), so they cannot both be true in the same situation (assuming at least two people). ■

Solution 4.9. Entailment: suppose $\exists x \forall y P(x, y)$, with witness x_0 . Given any y , $P(x_0, y)$ holds, so $\exists x P(x, y)$ holds; as y was arbitrary, $\forall y \exists x P(x, y)$. Converse fails: over $\mathbb{R}$ with $P(x, y)$: “ $x + y = 0$ ”, $\forall y \exists x$ is true ($x = -y$) but no single x works for every y , so $\exists x \forall y$ is false. ■

Rules of Inference

Notes By Pr. El Hadiq Zouhair

5.1 Motivation

An *argument* is a sequence of propositions: *premises* $p_1, \dots, p_k$ followed by a *conclusion* q . The argument is *valid* when the premises entail the conclusion: $(p_1 \wedge \dots \wedge p_k) \rightarrow q$ is a tautology. Validity concerns form, not content — a valid argument with true premises is called *sound* and its conclusion must be true.

5.1.1 The limits of truth tables

Validity can in principle be checked by a truth table, but an argument with n atomic propositions requires 2^n rows: 20 atoms already mean more than a million rows. Rules of inference replace this exponential check by short chains of certified elementary steps — the same economy that proofs provide throughout mathematics.

5.2 The rules of inference

Each rule below is a small valid argument, verified once by a truth table and then reused forever. We write premises above the line, conclusion below.

Rule	Premises	Conclusion	Tautology
Modus ponens	$p, \quad p \rightarrow q$	q	$(p \wedge (p \rightarrow q)) \rightarrow q$
Modus tollens	$\neg q, \quad p \rightarrow q$	$\neg p$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$
Hypothetical syllogism	$p \rightarrow q, \quad q \rightarrow r$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
Disjunctive syllogism	$p \vee q, \quad \neg p$	q	$((p \vee q) \wedge \neg p) \rightarrow q$
Addition	p	$p \vee q$	$p \rightarrow (p \vee q)$
Simplification	$p \wedge q$	p	$(p \wedge q) \rightarrow p$
Conjunction	$p, \quad q$	$p \wedge q$	$((p) \wedge (q)) \rightarrow (p \wedge q)$
Resolution	$p \vee q, \quad \neg p \vee r$	$q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$

Modus ponens is the engine of deduction; resolution alone underlies automated theorem provers and SAT solvers. Beware the two fallacies of Chapter 2: affirming the consequent (q and $p \rightarrow q$, hence p) and denying the antecedent ($\neg p$ and $p \rightarrow q$, hence $\neg q$) are not valid rules.

Example 5.1. Premises: “If it rains, the match is cancelled” ($r \rightarrow c$); “If the match is cancelled, the tickets are refunded” ($c \rightarrow t$); “The tickets are not refunded” ($\neg t$). Conclusion: “It does not rain” ($\neg r$).

Step	Formula	Justification
1	$r \rightarrow c$	premise
2	$c \rightarrow t$	premise
3	$r \rightarrow t$	hypothetical syllogism, 1, 2
4	$\neg t$	premise
5	$\neg r$	modus tollens, 3, 4

Example 5.2. Premises: $p \vee q, \quad p \rightarrow r, \quad q \rightarrow s, \quad \neg r$. Show: s .

Step	Formula	Justification
1	$p \rightarrow r$	premise
2	$\neg r$	premise
3	$\neg p$	modus tollens, 1, 2
4	$p \vee q$	premise
5	q	disjunctive syllogism, 4, 3
6	$q \rightarrow s$	premise
7	s	modus ponens, 6, 5

5.3 Rules for quantified statements

Four rules connect quantifiers with concrete instances. Here c denotes an element of the domain.

Rule	From	Infer
Universal instantiation (UI)	$\forall x P(x)$	$P(c)$ for any c
Universal generalisation (UG)	$P(c)$ for <i>arbitrary</i> c	$\forall x P(x)$
Existential instantiation (EI)	$\exists x P(x)$	$P(c)$ for some <i>fresh</i> name c
Existential generalisation (EG)	$P(c)$ for some c	$\exists x P(x)$

Two safety conditions: in UG, c must be arbitrary — no special assumption may have been made about it; in EI, the witness name c must be new, not previously used in the proof. Violating either produces plausible-looking nonsense.

Example 5.3 (Socrates). Premises: $\forall x (M(x) \rightarrow D(x))$ (“all men are mortal”) and $M(\text{socrates})$. By UI, $M(\text{socrates}) \rightarrow D(\text{socrates})$; by modus ponens, $D(\text{socrates})$: Socrates is mortal. Propositional logic alone could never perform the first step.

Example 5.4 (Existential reasoning). Premises: “Some student in the class has not done the homework” $\exists x (S(x) \wedge \neg H(x))$; “Every student in the class passed the test” $\forall x (S(x) \rightarrow T(x))$. Conclusion: “Someone who passed the test has not done the homework” $\exists x (T(x) \wedge \neg H(x))$. Proof: by EI choose a fresh c with $S(c) \wedge \neg H(c)$; by simplification $S(c)$ and $\neg H(c)$; by UI on the second premise, $S(c) \rightarrow T(c)$; by modus ponens $T(c)$; by conjunction $T(c) \wedge \neg H(c)$; by EG, $\exists x (T(x) \wedge \neg H(x))$.

5.4 Summary

Valid argument forms — modus ponens, modus tollens, syllogisms, addition, simplification, conjunction, resolution — replace exponential truth-table checks by linear chains of certified steps. The four quantifier rules (UI, UG, EI, EG) let deductions pass between general statements and concrete instances, with the side conditions “arbitrary” (UG) and “fresh” (EI). Together these rules constitute the deductive machinery used implicitly by every proof in the remainder of the book.

Exercises

Exercise 5.1 EASY

Name the rule of inference used: (a) “It is snowing, therefore it is snowing or it is cold.” (b) “It is snowing and cold; therefore it is cold.” (c) “If I sleep, I miss the bus. I slept. Therefore I missed the bus.”

Exercise 5.2 EASY

Verify by truth table that disjunctive syllogism $((p \vee q) \wedge \neg p) \rightarrow q$ is a tautology.

Exercise 5.3 EASY

Identify the fallacy: “If the program is correct, all tests pass. All tests pass. Therefore the program is correct.”

Exercise 5.4 MEDIUM

Construct a step-by-step deduction. Premises: $\neg p \rightarrow q$, $q \rightarrow r$, $\neg r$. Conclusion: p .

Exercise 5.5 MEDIUM

Construct a deduction. Premises: $p \rightarrow (q \wedge r)$, $p \vee s$, $\neg q$. Conclusion: s .

Exercise 5.6 MEDIUM

Use resolution repeatedly to derive a contradiction (the empty clause) from the clauses: $p \vee q$, $\neg p \vee q$, $\neg q$.

Exercise 5.7 MEDIUM

Premises: “Every computer science student studies discrete mathematics” and “Hind is a computer science student”. Formalise with predicates and derive “Hind studies discrete mathematics”, naming each rule.

Exercise 5.8 HARD

Find the flaw: from $\exists x P(x)$ and $\exists x Q(x)$, a student instantiates both with the same c to get $P(c) \wedge Q(c)$ and concludes $\exists x (P(x) \wedge Q(x))$. Exhibit concrete P , Q and a domain where the premises are true and the conclusion false.

Exercise 5.9 HARD

Premises: $\forall x (P(x) \rightarrow Q(x))$ and $\forall x (Q(x) \rightarrow R(x))$. Give a full quantified deduction of $\forall x (P(x) \rightarrow R(x))$, naming every rule (UI, UG, hypothetical syllogism).

Solutions to the Exercises

Solution 5.1. (a) Addition. (b) Simplification. (c) Modus ponens. ■

Solution 5.2. The formula fails only if $(p \vee q) \wedge \neg p$ is true and q false. Then p is false (from $\neg p$) and q false, making $p \vee q$ false — contradiction. So no falsifying row exists: a tautology. (The four-row table confirms: rows (T, T), (T, F) make $\neg p$ false; row (F, T) makes q true; row (F, F) makes $p \vee q$ false.) ■

Solution 5.3. Affirming the consequent: from $p \rightarrow q$ and q one cannot infer p — tests can pass for an incorrect program. ■

Solution 5.4. 1. $q \rightarrow r$ (premise); 2. $\neg r$ (premise); 3. $\neg q$ (modus tollens, 1,2); 4. $\neg p \rightarrow q$ (premise); 5. $\neg\neg p$, i.e. p (modus tollens, 4,3, with double negation). ■

Solution 5.5. 1. $\neg q$ (premise); 2. $\neg q \vee \neg r$ (addition, 1); 3. $\neg(q \wedge r)$ (De Morgan, 2); 4. $p \rightarrow (q \wedge r)$ (premise); 5. $\neg p$ (modus tollens, 4,3); 6. $p \vee s$ (premise); 7. s (disjunctive syllogism, 6,5). ■

Solution 5.6. Resolve $p \vee q$ with $\neg p \vee q$ on p : get $q \vee q = q$. Resolve q with $\neg q$: the empty clause. The clause set is unsatisfiable. ■

Solution 5.7. Let $C(x)$: “ x is a CS student”, $D(x)$: “ x studies discrete mathematics”. Premises: $\forall x (C(x) \rightarrow D(x))$, $C(\text{hind})$. 1. UI: $C(\text{hind}) \rightarrow D(\text{hind})$. 2. Modus ponens with the second premise: $D(\text{hind})$. ■

Solution 5.8. The error is reusing the name c in the second EI — the witness must be fresh. Counterexample: domain $\mathbb{Z}$, $P(x)$: “ x is even”, $Q(x)$: “ x is odd”. Both premises are true, yet no integer is both even and odd, so $\exists x (P(x) \wedge Q(x))$ is false. ■

Solution 5.9. Let c be arbitrary. 1. UI on premise 1: $P(c) \rightarrow Q(c)$. 2. UI on premise 2: $Q(c) \rightarrow R(c)$. 3. Hypothetical syllogism: $P(c) \rightarrow R(c)$. 4. Since c was arbitrary, UG yields $\forall x (P(x) \rightarrow R(x))$. ■

Part II

Discrete Structures

Sets

Notes By Pr. El Hadiq Zouhair

6.1 Definition

Definition 6.1 (Set). A *set* is an unordered collection of distinct objects, called its *elements* (or members). We write $a \in A$ when a is an element of A , and $a \notin A$ otherwise.

Sets can be described by *extension* (listing): $A = \{1, 2, 3, 5, 8\}$, or by *comprehension* (a defining property): $A = \{x \in \mathbb{N} \mid x \text{ is a Fibonacci number} \leq 8\}$. Order and repetition are irrelevant: $\{1, 2, 2, 3\} = \{3, 2, 1\}$.

Definition 6.2 (Equality). Two sets are equal when they have exactly the same elements: $A = B \iff \forall x (x \in A \leftrightarrow x \in B)$ (the *extensionality* principle).

6.2 Cardinality

Definition 6.3. If A contains exactly n distinct elements ($n \in \mathbb{N}$), A is *finite* and its *cardinality* is $|A| = n$. Otherwise A is *infinite*.

Example 6.4. $|\{a, b, c\}| = 3$; $|\emptyset| = 0$; $|\{\emptyset\}| = 1$ (a set containing one element, namely the empty set); $\mathbb{Z}$ is infinite.

6.3 Important sets

$$\mathbb{N} = \{0, 1, 2, \dots\}, \quad \mathbb{Z} = \{\dots, -1, 0, 1, \dots\}, \quad \mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}, \quad \mathbb{R}, \quad \mathbb{C}.$$

We write $\mathbb{Z}^+$ for the positive integers and $[a, b]$, (a, b) for real intervals. The chain $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ will be used constantly.

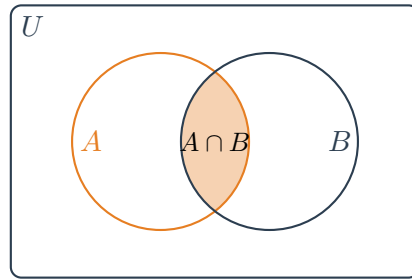
6.3.1 Countable and uncountable

Definition 6.5. A set is *countable* when it is finite or when its elements can be listed in an infinite sequence indexed by $\mathbb{N}$ (equivalently, when it has the same cardinality as $\mathbb{N}$, written $\aleph_0$). Otherwise it is *uncountable*.

$\mathbb{Z}$ is countable (list $0, 1, -1, 2, -2, \dots$) and so is $\mathbb{Q}$; Cantor's diagonal argument shows $\mathbb{R}$ is uncountable — there are strictly more reals than integers. Chapter 8 returns to cardinality via bijections.

6.4 Venn diagrams

A *Venn diagram* represents sets as regions (typically discs) inside a rectangle standing for the universal set U .



Venn diagrams guide intuition for the operations of Chapter 7, but a picture is not a proof: identities must be established by the element method or by membership tables.

6.5 Subsets

Definition 6.6. A is a *subset* of B , written $A \subseteq B$, when every element of A belongs to B : $\forall x (x \in A \rightarrow x \in B)$. If moreover $A \neq B$, A is a *proper subset*, written $A \subset B$.

Theorem 6.7. For every set A : $\emptyset \subseteq A$ and $A \subseteq A$. Moreover $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$ — the standard *double-inclusion* proof method.

Distinguish $\in$ from $\subseteq$: $1 \in \{1, 2\}$ but $1 \not\subseteq \{1, 2\}$; $\{1\} \subseteq \{1, 2\}$ and also $\{1\} \in \{\{1\}, 2\}$. Elements and subsets live at different levels.

6.6 The power set

Definition 6.8. The *power set* of A , written $\mathcal{P}(A)$, is the set of all subsets of A : $\mathcal{P}(A) = \{S \mid S \subseteq A\}$.

Example 6.9. $\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$; $\mathcal{P}(\emptyset) = \{\emptyset\}$.

Theorem 6.10. If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$.

Proof. A subset is determined by deciding, for each of the n elements, whether it is in or out: n independent binary choices, hence 2^n subsets by the product rule. (An induction proof appears in Chapter 13.) $\square$

6.7 Partitions

Definition 6.11. A *partition* of a set A is a family of nonempty, pairwise disjoint subsets (*blocks*) whose union is A : every element of A lies in exactly one block.

Example 6.12. $\{\{1, 3, 5, \dots\}, \{0, 2, 4, \dots\}\}$ — odds and evens — partitions $\mathbb{N}$ into two blocks. The sets $\{1, 2\}, \{2, 3\}$ do *not* partition $\{1, 2, 3\}$: the blocks overlap.

Partitions are exactly the structures induced by equivalence relations (Chapter 8) — this correspondence is one of the central theorems of discrete mathematics.

6.8 Cartesian products

Definition 6.13. The *ordered pair* (a, b) is determined by its components *and their order*: $(a, b) = (c, d) \iff a = c \wedge b = d$. The *Cartesian product* of A and B is

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

More generally $A_1 \times \dots \times A_n$ consists of all n -tuples.

Example 6.14. $\{1, 2\} \times \{a, b, c\} = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$, and $|A \times B| = |A| |B|$. Note $A \times B \neq B \times A$ in general. $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ is the coordinate plane.

6.9 Associations and datasets

Two structures built from sets pervade computing. An *association* (dictionary, map) is a finite set of key–value pairs $\{(k_1, v_1), \dots, (k_n, v_n)\}$ with distinct keys — formally a function from a finite key set (Chapter 8). A *dataset* (table) is a finite set or list of tuples drawn from a product $A_1 \times \dots \times A_n$: each row is a tuple, each column an attribute — precisely the data model of relational databases, built on the relations of Chapter 8.

6.10 The empty set

Definition 6.15. The *empty set* $\emptyset = \{\}$ is the unique set with no elements.

Every universally quantified statement about its elements is vacuously true; it is a subset of every set; and $|\emptyset| = 0$. Beware: $\emptyset \neq \{\emptyset\}$ — the latter has one element.

6.11 Summary

Sets are unordered collections of distinct elements, equal exactly when they have the same members. Key constructions: subsets and the subset order, the power set (2^n subsets for an n -element set), partitions (disjoint exhaustive blocks), and Cartesian products (ordered tuples). $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are countable; $\mathbb{R}$ is not. These objects are the raw material for all the structures to come.

Exercises

Exercise 6.1 EASY

List the elements of $A = \{x \in \mathbb{Z} \mid x^2 < 10\}$ and give $|A|$.

Exercise 6.2 EASY

True or false (justify briefly): (a) $\{1, 2, 3\} = \{3, 1, 2, 1\}$; (b) $\emptyset \in \{\emptyset\}$; (c) $\emptyset \subseteq \emptyset$; (d) $\{a\} \in \{a, \{a\}\}$ and $\{a\} \subseteq \{a, \{a\}\}$.

Exercise 6.3 EASY

Write out $\mathcal{P}(\{1, 2, 3\})$ and check its cardinality against Theorem 6.10.

Exercise 6.4 MEDIUM

Which of the following families are partitions of $\{1, 2, 3, 4, 5, 6\}$? (a) $\{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$; (b) $\{\{1, 2, 3\}, \{3, 4, 5, 6\}\}$; (c) $\{\{1, 4\}, \{2, 5\}\}$; (d) $\{\{1\}, \{2, 3, 4, 5, 6\}, \emptyset\}$.

Exercise 6.5 MEDIUM

Let $A = \{0, 1\}$. Compute $A \times A$, $A \times A \times A$ and $\mathcal{P}(A) \times A$, giving the cardinality of each.

Exercise 6.6 MEDIUM

Prove that $A \subseteq B$ if and only if $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Exercise 6.7 MEDIUM

Show that $\mathbb{Z}$ is countable by exhibiting an explicit enumeration, and give the formula of a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$ realising it.

Exercise 6.8 HARD

How many partitions does a 3-element set have? List them all for $\{a, b, c\}$. (The counts for growing n are the *Bell numbers*.)

Exercise 6.9 HARD

Prove that $(A \times B = B \times A)$ holds if and only if $A = B$, or $A = \emptyset$, or $B = \emptyset$.

Exercise 6.10 HARD

(Cantor, preview of Chapter 8.) Show that no function $f : A \rightarrow \mathcal{P}(A)$ is surjective, by considering $D = \{x \in A \mid x \notin f(x)\}$. Conclude that $|\mathcal{P}(A)| > |A|$ for every set A .

Solutions to the Exercises

Solution 6.1. $x^2 < 10$ for $x \in \{-3, -2, -1, 0, 1, 2, 3\}$, so A has these 7 elements: $|A| = 7$. ■

Solution 6.2. (a) True — repetition and order are irrelevant. (b) True — the set $\{\emptyset\}$ has $\emptyset$ as its single element. (c) True — every set is a subset of itself. (d) Both true: $\{a\}$ is listed as an element; and its only element a also belongs to $\{a, \{a\}\}$, giving the inclusion. ■

Solution 6.3. $\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$: $8 = 2^3$ subsets, as predicted. ■

Solution 6.4. (a) Yes: nonempty, disjoint, union is everything. (b) No: blocks overlap at 3. (c) No: union misses 3 and 6. (d) No: a partition may not contain the empty block. ■

Solution 6.5. $A \times A = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, $|A \times A| = 4$. A^3 has $2^3 = 8$ triples (x, y, z) with $x, y, z \in \{0, 1\}$. $\mathcal{P}(A) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$ has 4 elements, so $\mathcal{P}(A) \times A$ has $4 \times 2 = 8$ pairs (S, a) . ■

Solution 6.6. $(\Rightarrow)$ Assume $A \subseteq B$ and let $S \in \mathcal{P}(A)$, i.e. $S \subseteq A$. Then $S \subseteq A \subseteq B$, so $S \in \mathcal{P}(B)$. $(\Leftarrow)$ Assume $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Since $A \in \mathcal{P}(A)$, we get $A \in \mathcal{P}(B)$, i.e. $A \subseteq B$. ■

Solution 6.7. Enumerate $0, 1, -1, 2, -2, 3, -3, \dots$. Formally define $f : \mathbb{N} \rightarrow \mathbb{Z}$ by $f(n) = (n+1)/2$ if n is odd and $f(n) = -n/2$ if n is even. Each integer appears exactly once (positive k at $n = 2k - 1$, nonpositive $-k$ at $n = 2k$), so f is a bijection and $\mathbb{Z}$ is countable. ■

Solution 6.8. Five partitions: $\{\{a\}, \{b\}, \{c\}\}$; $\{\{a, b\}, \{c\}\}$; $\{\{a, c\}, \{b\}\}$; $\{\{b, c\}, \{a\}\}$; $\{\{a, b, c\}\}$. The Bell numbers begin $1, 1, 2, 5, 15, 52, \dots$. ■

Solution 6.9. If $A = B$ or either is empty, both products are equal (both empty in the latter case). Conversely suppose $A \times B = B \times A$ with both nonempty. Take any $a \in A$; pick $b \in B$. Then $(a, b) \in A \times B = B \times A$, so $a \in B$; hence $A \subseteq B$. Symmetrically $B \subseteq A$, so $A = B$. ■

Solution 6.10. Suppose $f : A \rightarrow \mathcal{P}(A)$ and let $D = \{x \in A \mid x \notin f(x)\} \in \mathcal{P}(A)$. If f were surjective, $D = f(d)$ for some $d \in A$. Then $d \in D \iff d \notin f(d) = D$ — a contradiction. So no f is surjective. The injection $a \mapsto \{a\}$ shows $|A| \leq |\mathcal{P}(A)|$; absence of a surjection rules out equality, hence $|\mathcal{P}(A)| > |A|$. ■

Operations on Sets

Notes By Pr. El Hadiq Zouhair

7.1 The basic operations

Throughout, sets live inside a fixed universal set U .

Definition 7.1 (Union, intersection, difference, complement).

$$\begin{aligned}
 A \cup B &= \{x \mid x \in A \vee x \in B\} && \text{(union)} \\
 A \cap B &= \{x \mid x \in A \wedge x \in B\} && \text{(intersection)} \\
 A \setminus B &= \{x \mid x \in A \wedge x \notin B\} && \text{(difference)} \\
 \overline{A} &= U \setminus A = \{x \in U \mid x \notin A\} && \text{(complement)} \\
 A \oplus B &= (A \setminus B) \cup (B \setminus A) && \text{(symmetric difference).}
 \end{aligned}$$

Two sets with $A \cap B = \emptyset$ are called *disjoint*.

Each set operation mirrors a logical connective: $\cup \leftrightarrow \vee$, $\cap \leftrightarrow \wedge$, $\overline{} \leftrightarrow \neg$, $\oplus \leftrightarrow \oplus$. This dictionary transfers every law of Chapter 2 into a set identity.

Example 7.2. With $U = \{1, \dots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$: $A \cup B = \{1, \dots, 7\}$, $A \cap B = \{4, 5\}$, $A \setminus B = \{1, 2, 3\}$, $\overline{B} = \{1, 2, 3, 8, 9, 10\}$, $A \oplus B = \{1, 2, 3, 6, 7\}$.

Theorem 7.3 (Inclusion–exclusion, basic form). For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$.

Example 7.4 (Combining operations). With the sets above, compute $\overline{A \cap B} \cap A$: $A \cap B = \{4, 5\}$, so $\overline{A \cap B} = U \setminus \{4, 5\}$, and intersecting with A leaves $\{1, 2, 3\} = A \setminus B$. In fact $\overline{A \cap B} \cap A = A \setminus B$ always — provable by the identities below.

Example 7.5 (Empty regions). $A \cap \overline{A} = \emptyset$ and $(A \setminus B) \cap (B \setminus A) = \emptyset$ for any A, B : some Venn regions are empty by pure logic, regardless of the particular sets.

7.2 Set identities

Theorem 7.6 (Set identities). For all $A, B, C \subseteq U$:

Identity	$A \cup \emptyset = A, \quad A \cap U = A$
Domination	$A \cup U = U, \quad A \cap \emptyset = \emptyset$
Idempotence	$A \cup A = A, \quad A \cap A = A$
Double complement	$\overline{\overline{A}} = A$
Commutativity	$A \cup B = B \cup A, \quad A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C),$ and dually
Distributivity	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$ and dually
De Morgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{A \cap B} = \overline{A} \cup \overline{B}$
Absorption	$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$
Complement	$A \cup \overline{A} = U, \quad A \cap \overline{A} = \emptyset$

Also $A \setminus B = A \cap \overline{B}$, the workhorse for eliminating differences.

7.3 Proving identities

Three standard methods.

1. The element method (double inclusion). To prove $X = Y$, show $x \in X \Rightarrow x \in Y$ and conversely, unfolding the definitions into logic and applying the propositional laws.

Example 7.7. Prove $\overline{A \cup B} = \overline{A} \cap \overline{B}$:

$$x \in \overline{A \cup B} \iff \neg(x \in A \vee x \in B) \iff (x \notin A) \wedge (x \notin B) \iff x \in \overline{A} \cap \overline{B},$$

the middle step being De Morgan's law for propositions.

2. Algebraic manipulation. Chain identities from Theorem 7.6, replacing equals by equals — as in Example 7.4: $\overline{A \cap B} \cap A = (\overline{A \cup B}) \cap A = (\overline{A} \cap \overline{B}) \cap A = \emptyset \cup (\overline{B} \cap A) = A \setminus B$.

3. Membership tables. Analogous to truth tables: each row records whether x belongs (1) or not (0) to each set; two expressions are equal iff their columns coincide.

Example 7.8 (Membership-table proof). $A \oplus B = (A \cup B) \setminus (A \cap B)$:

A	B	$A \cup B$	$A \cap B$	$(A \cup B) \setminus (A \cap B)$	$A \oplus B$
1	1	1	1	0	0
1	0	1	0	1	1
0	1	1	0	1	1
0	0	0	0	0	0

The last two columns agree, proving the identity.

A membership table over n set letters has 2^n rows — it is exactly a truth table in disguise, so the method is complete for identities built from $\cup, \cap, \setminus, \oplus$ and complement.

7.4 Generalised unions and intersections

For a family $A_1, \dots, A_n$ (or an infinite family):

$$\bigcup_{i=1}^n A_i = \{x \mid \exists i, x \in A_i\}, \quad \bigcap_{i=1}^n A_i = \{x \mid \forall i, x \in A_i\}.$$

Example 7.9. With $A_i = \{i, i+1, i+2, \dots\} \subseteq \mathbb{N}$: $\bigcup_{i \geq 1} A_i = \mathbb{N}$ and $\bigcap_{i \geq 1} A_i = \emptyset$ (no natural number belongs to every A_i).

7.5 Summary

Union, intersection, difference, complement and symmetric difference mirror $\vee, \wedge, \setminus, \neg, \oplus$; consequently the identity, domination, idempotent, distributive, De Morgan, absorption and complement laws all transfer from logic to sets. Identities are proved by the element method, by algebraic chaining, or mechanically by membership tables.

Exercises

Exercise 7.1 EASY

Let $U = \{1, \dots, 12\}$, $A = \{1, 3, 5, 7, 9, 11\}$, $B = \{3, 6, 9, 12\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$, $B \setminus A$, $\overline{A}$ and $A \oplus B$.

Exercise 7.2 EASY

Draw (or describe region by region) the Venn diagram of $(A \cup B) \setminus (A \cap B)$ and name the operation it represents.

Exercise 7.3 EASY

Two clubs have 28 and 19 members; 7 people belong to both. How many people belong to at least one club?

Exercise 7.4 MEDIUM

Prove by the element method: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Exercise 7.5 MEDIUM

Prove algebraically (citing identities): $A \cup (B \setminus A) = A \cup B$.

Exercise 7.6 MEDIUM

Verify by a membership table: $A \setminus (A \setminus B) = A \cap B$.

Exercise 7.7 MEDIUM

Show that $A \subseteq B$ if and only if $A \cup B = B$, and if and only if $A \cap B = A$.

Exercise 7.8 HARD

Prove that symmetric difference is associative: $(A \oplus B) \oplus C = A \oplus (B \oplus C)$. *Hint: a membership table over three letters, or the parity characterisation.*

Exercise 7.9 HARD

Solve for X : the equation $A \oplus X = B$ has exactly one solution for given $A, B \subseteq U$. Find it and prove uniqueness.

Exercise 7.10 HARD

Prove the three-set inclusion-exclusion formula:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Solutions to the Exercises

Solution 7.1. $A \cup B = \{1, 3, 5, 6, 7, 9, 11, 12\}$; $A \cap B = \{3, 9\}$; $A \setminus B = \{1, 5, 7, 11\}$; $B \setminus A = \{6, 12\}$; $\overline{A} = \{2, 4, 6, 8, 10, 12\}$; $A \oplus B = \{1, 5, 6, 7, 11, 12\}$. ■

Solution 7.2. The region consists of the two “lune” parts of the discs — everything in exactly one of A, B , excluding the central lens $A \cap B$ and the exterior. This is the symmetric difference $A \oplus B$. ■

Solution 7.3. $|A \cup B| = 28 + 19 - 7 = 40$. ■

Solution 7.4. $x \in A \setminus (B \cup C) \iff x \in A \wedge \neg(x \in B \vee x \in C) \iff x \in A \wedge x \notin B \wedge x \notin C$ (De Morgan). Also $x \in (A \setminus B) \cap (A \setminus C) \iff (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C)$, which by idempotence of $\wedge$ is the same condition. Hence the sets are equal. ■

Solution 7.5. $A \cup (B \setminus A) = A \cup (B \cap \overline{A}) = (A \cup B) \cap (A \cup \overline{A}) = (A \cup B) \cap U = A \cup B$, using difference-as-intersection, distributivity, complement and identity laws. ■

Solution 7.6.

A	B	$A \setminus B$	$A \setminus (A \setminus B)$	$A \cap B$
1	1	0	1	1
1	0	1	0	0
0	1	0	0	0
0	0	0	0	0

Columns 4 and 5 coincide. ■

Solution 7.7. If $A \subseteq B$: every element of $A \cup B$ is in B , so $A \cup B \subseteq B \subseteq A \cup B$, giving equality; and $A \subseteq A \cap B \subseteq A$ similarly. Conversely, if $A \cup B = B$ then $A \subseteq A \cup B = B$; if $A \cap B = A$ then $A = A \cap B \subseteq B$. ■

Solution 7.8. Membership in a symmetric difference holds exactly when membership holds in an *odd* number of the operands: for two sets this is the definition. Then $x \in (A \oplus B) \oplus C$ iff x lies in an odd number of $\{A, B, C\}$ — check: it requires oddness between $(A \oplus B)$ and C , and $x \in A \oplus B$ records oddness between A and B ; combining parities gives overall parity. The same characterisation holds for $A \oplus (B \oplus C)$, so the two sides are equal. (The eight-row membership table gives the same conclusion mechanically.) ■

Solution 7.9. Take $X = A \oplus B$. Then, using associativity (Exercise 7.8), $A \oplus (A \oplus B) = (A \oplus A) \oplus B = \emptyset \oplus B = B$. Uniqueness: if $A \oplus X = A \oplus Y$, then XOR-ing both sides with A gives $X = Y$. (The structure $(\mathcal{P}(U), \oplus)$ is a group in which every element is its own inverse.) ■

Solution 7.10. Count the contribution of an element x belonging to exactly $k \geq 1$ of the three sets. Left side counts x once. Right side counts it $\binom{k}{1} - \binom{k}{2} + \binom{k}{3}$ times: for $k = 1$: 1; $k = 2$: $2 - 1 = 1$; $k = 3$: $3 - 3 + 1 = 1$. Every element is counted once on each side, so the totals agree. (Chapter 10 generalises this to n sets.) ■

Relations and Functions

Notes By Pr. El Hadiq Zouhair

8.1 Relations

8.1.1 Definition and representations

Definition 8.1. A *binary relation* from A to B is a subset $R \subseteq A \times B$. We write $a R b$ for $(a, b) \in R$. When $A = B$, R is a relation *on* A . An *n -ary relation* on $A_1, \dots, A_n$ is a subset of $A_1 \times \dots \times A_n$.

Example 8.2. On $A = \{1, 2, 3, 4\}$, the divisibility relation $R = \{(a, b) \mid a \mid b\}$ is $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.

A relation admits three equivalent representations: as a *set of ordered pairs*; as a *Boolean matrix* M_R with $(M_R)_{ij} = 1$ iff $(a_i, b_j) \in R$; and, when $A = B$, as a *directed graph* on vertex set A with an arc $a \rightarrow b$ for each $(a, b) \in R$. Matrix and graph views turn abstract questions into visual or computational ones: e.g. reflexivity is a diagonal of 1s; symmetry is $M_R = M_R^T$.

Being sets, relations combine by $\cup, \cap, \setminus$; moreover the *inverse* $R^{-1} = \{(b, a) \mid (a, b) \in R\}$ and the *composition* $S \circ R = \{(a, c) \mid \exists b, (a, b) \in R \wedge (b, c) \in S\}$ produce new relations from old.

8.1.2 Properties of relations on a set

Definition 8.3. A relation R on A is

- *reflexive* if $\forall a, (a, a) \in R$;
- *symmetric* if $(a, b) \in R \Rightarrow (b, a) \in R$;
- *antisymmetric* if $(a, b) \in R \wedge (b, a) \in R \Rightarrow a = b$;
- *asymmetric* if $(a, b) \in R \Rightarrow (b, a) \notin R$;
- *transitive* if $(a, b) \in R \wedge (b, c) \in R \Rightarrow (a, c) \in R$.

Symmetric and antisymmetric are not opposites: the equality relation is both; the relation $\{(1, 2), (2, 1), (1, 3)\}$ is neither. Asymmetric = antisymmetric + irreflexive.

Example 8.4. On $\mathbb{Z}$: $\leq$ is reflexive, antisymmetric, transitive; $<$ is asymmetric and transitive; “ $a \equiv b \pmod{5}$ ” is reflexive, symmetric, transitive; “ a divides b ” on $\mathbb{Z}^+$ is reflexive, antisymmetric, transitive.

8.1.3 Equivalence relations

Definition 8.5. An *equivalence relation* is a reflexive, symmetric and transitive relation on A . The *equivalence class* of a is $[a] = \{x \in A \mid x R a\}$; the set of classes is the *quotient* A/R .

Theorem 8.6 (Fundamental partition theorem). The equivalence classes of an equivalence relation on A form a partition of A ; conversely, every partition of A arises this way from exactly one equivalence relation (“belong to the same block”).

Proof. Reflexivity puts each a in $[a]$, so classes are nonempty and cover A . Suppose $[a] \cap [b] \neq \emptyset$, say c in both. For any $x \in [a]$: $x R a$, $a R c$ (symmetry from $c R a$), $c R b$, so by transitivity $x R b$, i.e. $[a] \subseteq [b]$; symmetrically $[b] \subseteq [a]$. Thus classes are equal or disjoint. The converse verification is routine. $\square$

Example 8.7 (Congruence modulo 3). On $\mathbb{Z}$ define $a \sim b \iff 3 \mid (a - b)$. Reflexive: $3 \mid 0$. Symmetric: $3 \mid (a - b) \Rightarrow 3 \mid (b - a)$. Transitive: $3 \mid (a - b)$, $3 \mid (b - c) \Rightarrow 3 \mid ((a - b) + (b - c)) = a - c$. The classes are $[0] = \{\dots, -3, 0, 3, \dots\}$, $[1] = \{\dots, -2, 1, 4, \dots\}$, $[2] = \{\dots, -1, 2, 5, \dots\}$ — the partition of $\mathbb{Z}$ into residues mod 3, and $\mathbb{Z}/\sim = \mathbb{Z}_3$.

8.1.4 Order relations

Definition 8.8. A *partial order* is a reflexive, antisymmetric, transitive relation (e.g. $\subseteq$ on $\mathcal{P}(S)$, $\mid$ on $\mathbb{Z}^+$). It is a *total order* when any two elements are comparable (e.g. $\leq$ on $\mathbb{R}$). A *strict order* (e.g. $<$, $\subset$) is irreflexive and transitive.

8.1.5 Closures

Definition 8.9. The *reflexive* (resp. *symmetric*, *transitive*) *closure* of R is the smallest relation containing R with the stated property: $r(R) = R \cup \{(a, a)\}_{a \in A}$, $s(R) = R \cup R^{-1}$, and $t(R) = \bigcup_{k \geq 1} R^k$, where R^k is the k -fold composition — $(a, b) \in t(R)$ iff the digraph of R has a path from a to b .

8.2 Functions

8.2.1 Definition

Definition 8.10. A function $f : A \rightarrow B$ is a relation $f \subseteq A \times B$ such that every $a \in A$ is related to *exactly one* $b \in B$, written $b = f(a)$. A is the *domain*, B the *codomain*, and $f(A) = \{f(a) \mid a \in A\}$ the *range* (image).

Useful special cases: the *identity* $\text{id}_A(a) = a$; *constant* functions; the *restriction* $f|_S : S \rightarrow B$ of f to $S \subseteq A$; and *extensions*, which enlarge the domain.

8.2.2 Injections, surjections, bijections

Definition 8.11. $f : A \rightarrow B$ is

- *injective* (one-to-one) if $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$ — distinct inputs give distinct outputs;
- *surjective* (onto) if $\forall b \in B \exists a \in A, f(a) = b$ — the range is all of B ;
- *bijective* if it is both.

Example 8.12. $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 2x$ is injective, not surjective. $g : \mathbb{R} \rightarrow [0, \infty), g(x) = x^2$ is surjective, not injective. $h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = x^3$ is bijective.

Theorem 8.13 (Finite-case shortcut). Let $|A| = |B|$ be finite and $f : A \rightarrow B$. Then f injective $\iff f$ surjective $\iff f$ bijective.

This equivalence — a form of the pigeonhole principle (Chapter 10) — fails for infinite sets: $x \mapsto 2x$ on $\mathbb{Z}$ is injective but not surjective.

8.2.3 Composition

Definition 8.14. The *composition* of $f : A \rightarrow B$ and $g : B \rightarrow C$ is $g \circ f : A \rightarrow C$, $(g \circ f)(a) = g(f(a))$.

Theorem 8.15. Composition is associative. If f and g are injective (resp. surjective, bijective), so is $g \circ f$. Conversely, if $g \circ f$ is injective then f is injective; if $g \circ f$ is surjective then g is surjective.

8.2.4 Inverse functions

Theorem 8.16 (Existence of the inverse). $f : A \rightarrow B$ has an *inverse function* $f^{-1} : B \rightarrow A$ (with $f^{-1}(f(a)) = a$ and $f(f^{-1}(b)) = b$) if and only if f is a bijection. Moreover $(f^{-1})^{-1} = f$ and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof sketch. If f is bijective, each $b \in B$ has exactly one preimage, and sending b to it defines f^{-1} . If f^{-1} exists, f must be injective (two equal images would need two values of f^{-1}) and surjective (every b equals $f(f^{-1}(b))$). $\square$

8.2.5 Images and preimages of sets

Definition 8.17. For $S \subseteq A$ and $T \subseteq B$: $f(S) = \{f(x) \mid x \in S\}$ and $f^{-1}(T) = \{x \in A \mid f(x) \in T\}$ (defined even when f has no inverse function).

Preimages behave perfectly: $f^{-1}(T_1 \cup T_2) = f^{-1}(T_1) \cup f^{-1}(T_2)$ and likewise for $\cap$ and complements. Images preserve unions, but only $f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2)$ in general, with equality for injective f .

8.3 Cardinality via bijections

Definition 8.18. Sets A and B are *equinumerous*, $|A| = |B|$, when a bijection $A \rightarrow B$ exists. A is countable when $|A| \leq |\mathbb{N}|$.

Example 8.19 (A bijection $(0, 1) \rightarrow \mathbb{R}$). $x \mapsto \tan(\pi x - \frac{\pi}{2})$ maps $(0, 1)$ bijectively onto $\mathbb{R}$ (strictly increasing, range $\mathbb{R}$). Hence a bounded interval has *exactly as many* points as the whole real line.

Theorem 8.20 (Cantor). For every set A , $|A| < |\mathcal{P}(A)|$: the diagonal set $D = \{x \in A \mid x \notin f(x)\}$ witnesses that no $f : A \rightarrow \mathcal{P}(A)$ is surjective.

Consequently there is an infinite hierarchy of infinities: $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| = |\mathbb{R}| < |\mathcal{P}(\mathbb{R})| < \dots$

8.4 Summary

Relations are subsets of products, viewable as pair-sets, Boolean matrices or digraphs; the key property bundles are *equivalence relations* (= partitions, Theorem 8.6) and *orders*. Functions are total, single-valued relations; injectivity, surjectivity and bijectivity govern composition (Theorem 8.15) and invertibility (Theorem 8.16); bijections define cardinality, and Cantor's theorem shows the power set always strictly exceeds the set.

Exercises

Exercise 8.1 EASY

Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$. Decide whether R is reflexive, symmetric, antisymmetric, transitive.

Exercise 8.2 EASY

Write the Boolean matrix of $R = \{(a, b) \in A^2 \mid a < b\}$ for $A = \{1, 2, 3, 4\}$, and describe its digraph.

Exercise 8.3 EASY

Which of these functions $\mathbb{Z} \rightarrow \mathbb{Z}$ are injective? surjective? (a) $f(n) = n + 3$; (b) $g(n) = n^2$; (c) $h(n) = \lceil n/2 \rceil$.

Exercise 8.4 MEDIUM

Prove that congruence modulo m ($m \geq 1$) is an equivalence relation on $\mathbb{Z}$, and determine the number of equivalence classes.

Exercise 8.5 MEDIUM

On $\mathbb{R}^2$, define $(x_1, y_1) \sim (x_2, y_2) \iff x_1^2 + y_1^2 = x_2^2 + y_2^2$. Show $\sim$ is an equivalence relation and describe its classes geometrically.

Exercise 8.6 MEDIUM

Show that divisibility $|$ is a partial order on $\mathbb{Z}^+$ but not a total order. Exhibit two incomparable elements.

Exercise 8.7 MEDIUM

Let $R = \{(1, 2), (2, 3), (3, 1)\}$ on $A = \{1, 2, 3\}$. Compute the reflexive, symmetric and transitive closures of R .

Exercise 8.8 MEDIUM

Let $f(x) = 3x - 4$ on $\mathbb{R}$. Show f is a bijection and find f^{-1} . Verify $(f^{-1} \circ f)(x) = x$.

Exercise 8.9 MEDIUM

Give an example of $f, g : \mathbb{N} \rightarrow \mathbb{N}$ such that $g \circ f$ is injective but g is not, and explain which halves of Theorem 8.15 this illustrates.

Exercise 8.10 HARD

Let $f : A \rightarrow B$ and $S_1, S_2 \subseteq A$. Prove $f(S_1 \cap S_2) \subseteq f(S_1) \cap f(S_2)$, show equality can fail, and prove equality holds for all S_1, S_2 iff f is injective.

Exercise 8.11 HARD

Construct an explicit bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ and deduce that $\mathbb{Q}^+$ is countable.

Exercise 8.12 HARD

Using Theorem 8.13, prove: if $|A| = |B| = n$ and $f : A \rightarrow B$ is injective, then f is bijective. Then give a counterexample for $A = B = \mathbb{N}$.

Solutions to the Exercises

Solution 8.1. Reflexive: yes — $(1, 1), (2, 2), (3, 3) \in R$. Symmetric: yes — the only non-diagonal pairs $(1, 2), (2, 1)$ come in mirror pairs. Antisymmetric: no — $(1, 2)$ and $(2, 1)$ with $1 \neq 2$. Transitive: yes — all required compositions land in R (e.g. $(1, 2), (2, 1) \Rightarrow (1, 1) \in R$). ■

Solution 8.2. $M_R = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ — strictly upper triangular. The digraph has arcs from each smaller number to each larger one and no loops: a transitive tournament-like DAG. ■

Solution 8.3. (a) Bijective: injective since $n + 3 = m + 3 \Rightarrow n = m$; surjective since $n = k - 3$ maps to any k . (b) Neither: $g(-1) = g(1)$ kills injectivity; negative integers are not squares. (c) Surjective: every k is $h(2k)$; not injective: $h(1) = h(2) = 1$. ■

Solution 8.4. Reflexive: $m \mid 0$. Symmetric: $m \mid (a - b) \Rightarrow m \mid (b - a)$. Transitive: $m \mid (a - b)$, $m \mid (b - c)$ imply $m \mid (a - c)$ (sum). The classes are $[0], [1], \dots, [m - 1]$, one per remainder: exactly m classes. ■

Solution 8.5. The relation compares values of $\varphi(x, y) = x^2 + y^2$; “same value under a function” is always reflexive, symmetric, transitive. The class of (a, b) is the circle centred at the origin through (a, b) (the origin alone for radius 0): the plane is partitioned into concentric circles. ■

Solution 8.6. Reflexive: $a \mid a$. Antisymmetric: $a \mid b$ and $b \mid a$ with $a, b > 0$ force $a = b$. Transitive: $a \mid b$ and $b \mid c \Rightarrow a \mid c$. Not total: $2 \nmid 3$ and $3 \nmid 2$, so 2 and 3 are incomparable. ■

Solution 8.7. $r(R) = R \cup \{(1, 1), (2, 2), (3, 3)\}$. $s(R) = R \cup \{(2, 1), (3, 2), (1, 3)\}$. For $t(R)$: paths give $(1, 3)$ via 2, $(2, 1)$ via 3, $(3, 2)$ via 1, then $(1, 1), (2, 2), (3, 3)$ via three-step cycles: $t(R) = A \times A$. ■

Solution 8.8. Injective: $3x - 4 = 3y - 4 \Rightarrow x = y$. Surjective: given y , $x = (y + 4)/3$ satisfies $f(x) = y$. Hence bijective with $f^{-1}(y) = (y + 4)/3$, and $f^{-1}(f(x)) = ((3x - 4) + 4)/3 = x$. ■

Solution 8.9. Let $f(n) = 2n$ and let g map $2k \mapsto k$ and $2k + 1 \mapsto 0$. Then $(g \circ f)(n) = n$ is injective (indeed bijective), yet $g(1) = g(3) = 0$ is not injective. This matches Theorem 8.15: injectivity of $g \circ f$ forces only f to be injective, and surjectivity of $g \circ f$ forces only g to be surjective (g here is surjective). ■

Solution 8.10. If $y \in f(S_1 \cap S_2)$, then $y = f(x)$ with $x \in S_1$ and $x \in S_2$, so $y \in f(S_1)$ and $y \in f(S_2)$. Failure: $f(x) = x^2$, $S_1 = \{-1\}$, $S_2 = \{1\}$: $f(S_1 \cap S_2) = f(\emptyset) = \emptyset$ but $f(S_1) \cap f(S_2) = \{1\}$. If f is injective and $y \in f(S_1) \cap f(S_2)$, then $y = f(x_1) = f(x_2)$ with $x_i \in S_i$; injectivity gives $x_1 = x_2 \in S_1 \cap S_2$, so $y \in f(S_1 \cap S_2)$. Conversely, if f is not injective, the counterexample pattern above (two points with equal image) breaks equality. ■

Solution 8.11. Cantor pairing: $\pi(m, n) = \frac{(m+n)(m+n+1)}{2} + n$ enumerates the diagonals $m + n = 0, 1, 2, \dots$ in order; each pair receives a distinct index and every index is reached: a bijection $\mathbb{N}^2 \rightarrow \mathbb{N}$. Each positive rational is p/q with $(p, q) \in \mathbb{N}^2$ (coprime, $q \geq 1$), so $\mathbb{Q}^+$ injects into $\mathbb{N}^2 \sim \mathbb{N}$; being infinite, it is countable. ■

Solution 8.12. If f were not surjective, its range would have at most $n - 1$ elements, and $f : A \rightarrow f(A)$ with $|A| = n > |f(A)|$ would force two elements to share an image (pigeonhole), contradicting injectivity. Hence f is surjective, so bijective. For $A = B = \mathbb{N}$, $f(n) = n + 1$ is injective but misses 0: the shortcut is strictly a finite phenomenon. ■

Sequences and Series

Notes By Pr. El Hadiq Zouhair

9.1 Sequences

Definition 9.1. A *sequence* in a set X is a function $a : \mathbb{N} \rightarrow X$ (or $\mathbb{Z}^+ \rightarrow X$); we write a_n for $a(n)$ and $(a_n)_{n \geq 0}$ for the sequence. A *subsequence* $(a_{n_k})_{k \geq 0}$ is obtained by selecting indices $n_0 < n_1 < n_2 < \dots$.

Definition 9.2. A real sequence (a_n) is *bounded* if $\exists M, |a_n| \leq M$ for all n ; *increasing* if $a_{n+1} \geq a_n$ for all n (strictly, with $>$); *decreasing* dually; *monotone* if either.

9.2 Limits

Definition 9.3 (Limit). (a_n) *converges* to $L \in \mathbb{R}$, written $\lim_{n \rightarrow \infty} a_n = L$, when

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, |a_n - L| < \varepsilon.$$

A sequence with no limit *diverges*.

Example 9.4. $a_n = \frac{1}{n} \rightarrow 0$: given $\varepsilon > 0$ take $N > 1/\varepsilon$; then $n \geq N \Rightarrow |1/n| \leq 1/N < \varepsilon$.

Theorem 9.5 (Uniqueness). A sequence has at most one limit.

Proof. If $a_n \rightarrow L$ and $a_n \rightarrow L'$ with $L \neq L'$, take $\varepsilon = |L - L'|/2$. Beyond the larger of the two thresholds, $|L - L'| \leq |L - a_n| + |a_n - L'| < 2\varepsilon = |L - L'|$ — contradiction. $\square$

Theorem 9.6 (Algebra of limits). If $a_n \rightarrow A$ and $b_n \rightarrow B$ then $a_n + b_n \rightarrow A + B$, $a_n b_n \rightarrow AB$, and $a_n/b_n \rightarrow A/B$ provided $B \neq 0$.

Theorem 9.7 (Monotone convergence). Every increasing sequence bounded above converges (to its supremum); every decreasing sequence bounded below converges.

Theorem 9.8 (Squeeze). If $a_n \leq b_n \leq c_n$ eventually and $a_n \rightarrow L$, $c_n \rightarrow L$, then $b_n \rightarrow L$.

Definition 9.9. (a_n) is *Cauchy* when $\forall \varepsilon > 0 \exists N \forall m, n \geq N, |a_m - a_n| < \varepsilon$.

In $\mathbb{R}$, a sequence converges iff it is Cauchy (completeness), and every bounded sequence has a convergent subsequence (Bolzano–Weierstrass).

9.2.1 Standard limits

$$\frac{1}{n^p} \rightarrow 0 \ (p > 0); \quad r^n \rightarrow 0 \ (|r| < 1); \quad n^{1/n} \rightarrow 1; \quad \left(1 + \frac{x}{n}\right)^n \rightarrow e^x; \quad \frac{n!}{n^n} \rightarrow 0.$$

9.3 Progressions and finite sums

Definition 9.10 (AP). An *arithmetic progression* has constant difference: $a_n = a_0 + nd$. The sum of the first n terms is

$$S_n = \sum_{k=0}^{n-1} (a_0 + kd) = na_0 + d \frac{n(n-1)}{2} = \frac{n}{2} (\text{first} + \text{last}).$$

Definition 9.11 (GP). A *geometric progression* has constant ratio: $a_n = a_0 r^n$. For $r \neq 1$,

$$\sum_{k=0}^{n-1} a_0 r^k = a_0 \frac{1 - r^n}{1 - r}, \quad \text{and for } |r| < 1, \quad \sum_{k=0}^{\infty} a_0 r^k = \frac{a_0}{1 - r}.$$

Theorem 9.12 (Standard closed forms).

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Proof of the second formula, by induction. Base $n = 1$: $1 = \frac{1 \cdot 2 \cdot 3}{6}$. Step: assuming the formula for n ,

$$\sum_{k=1}^{n+1} k^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n+1)(2n^2 + n + 6n + 6)}{6} = \frac{(n+1)(n+2)(2n+3)}{6},$$

which is the formula at $n + 1$. (Induction is treated fully in Chapter 13.) $\square$

9.4 Series

Definition 9.13. Given $(a_n)_{n \geq 1}$, the *series* $\sum_{n \geq 1} a_n$ is the sequence of *partial sums* $S_N = \sum_{n=1}^N a_n$; the series *converges* to S when $S_N \rightarrow S$.

Theorem 9.14 (Necessary condition). If $\sum a_n$ converges then $a_n \rightarrow 0$. The converse fails: the *harmonic series* $\sum 1/n$ diverges although $1/n \rightarrow 0$.

Divergence of the harmonic series. Group terms: $\frac{1}{2} \geq \frac{1}{2}$; $\frac{1}{3} + \frac{1}{4} > \frac{2}{4} = \frac{1}{2}$; $\frac{1}{5} + \dots + \frac{1}{8} > \frac{4}{8} = \frac{1}{2}$; each block of 2^k terms exceeds $\frac{1}{2}$, so $S_{2^m} > 1 + \frac{m}{2} \rightarrow \infty$. $\square$

Example 9.15 (Telescoping). $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N+1} \right) = 1$.

Theorem 9.16 (p -series). $\sum_{n \geq 1} n^{-p}$ converges iff $p > 1$.

9.4.1 Convergence tests

Theorem 9.17 (Tests for positive series). Let $a_n, b_n \geq 0$.

- **Comparison:** if $a_n \leq b_n$ eventually and $\sum b_n$ converges, then $\sum a_n$ converges; contrapositively divergence passes upward.
- **Ratio (d'Alembert):** if $a_{n+1}/a_n \rightarrow \ell$, the series converges for $\ell < 1$ and diverges for $\ell > 1$ ($\ell = 1$: inconclusive).
- **Root (Cauchy):** same conclusion with $\sqrt[n]{a_n} \rightarrow \ell$.
- **Integral:** for f positive decreasing, $\sum f(n)$ and $\int_1^{\infty} f(x) dx$ converge or diverge together.

Theorem 9.18 (Leibniz alternating test). If (b_n) decreases to 0, then $\sum (-1)^{n+1} b_n$ converges, and the error after N terms is at most b_{N+1} .

Definition 9.19. $\sum a_n$ is *absolutely convergent* when $\sum |a_n|$ converges; absolute convergence implies convergence. A convergent but not absolutely convergent series (e.g. $\sum (-1)^{n+1}/n$) is *conditionally convergent* — and by Riemann's theorem its terms can be rearranged to reach any prescribed sum.

9.5 Recurrence relations

Definition 9.20. A sequence is defined by a *recurrence* when each term is given as a function of earlier terms, together with initial conditions — e.g. the Fibonacci sequence $F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$.

Theorem 9.21 (Linear homogeneous recurrences, order 2). For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, form the characteristic equation $x^2 = c_1 x + c_2$. If it has distinct roots $r_1 \neq r_2$, the general solution is $a_n = \alpha r_1^n + \beta r_2^n$; if a double root r , then $a_n = (\alpha + \beta n) r^n$. The constants are fixed by the initial conditions.

Example 9.22 (Binet's formula). For Fibonacci, $x^2 = x + 1$ gives roots $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$. Solving $\alpha + \beta = 0$, $\alpha\varphi + \beta\psi = 1$ yields

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

Since $|\psi| < 1$, F_n is the nearest integer to $\varphi^n/\sqrt{5}$, and $F_{n+1}/F_n \rightarrow \varphi$, the golden ratio.

9.6 Products and famous sequences

The product notation mirrors the sum: $\prod_{k=1}^n a_k = a_1 a_2 \cdots a_n$; e.g. $n! = \prod_{k=1}^n k$, and Wallis' product $\prod_{k \geq 1} \frac{4k^2}{4k^2 - 1} = \frac{\pi}{2}$. Beyond Fibonacci, classical sequences include the Catalan numbers $C_n = \frac{1}{n+1} \binom{2n}{n}$ (Chapter 11), the harmonic numbers $H_n = \sum_{k \leq n} 1/k \approx \ln n + \gamma$, the primes, and the triangular numbers $n(n+1)/2$.

9.7 Summary

Sequences are functions on $\mathbb{N}$; convergence is the ε - N game, tamed by uniqueness, limit algebra, monotone convergence, squeeze and Cauchy criteria. AP/GP sums and the power sums of Theorem 9.12 are the basic closed forms. Series converge through comparison, ratio, root, integral and Leibniz tests; absolute convergence is the robust notion. Linear recurrences are solved by characteristic roots — Binet's formula being the showcase.

Exercises

Exercise 9.1 EASY

Write the first five terms of: (a) $a_n = \frac{(-1)^n}{n+1}$ ($n \geq 0$); (b) the sequence defined by $b_0 = 2$, $b_n = 3b_{n-1} - 1$.

Exercise 9.2 EASY

An amphitheatre has 20 rows; the first has 18 seats and each row has 3 more than the previous. How many seats in total?

Exercise 9.3 EASY

Compute $\sum_{k=0}^9 5 \cdot 2^k$ and $\sum_{k=0}^{\infty} 5 \cdot (1/3)^k$.

Exercise 9.4 MEDIUM

Prove from Definition 9.3 that $\lim_{n \rightarrow \infty} \frac{2n+1}{n+3} = 2$.

Exercise 9.5 MEDIUM

Let $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$. Show (a_n) is increasing and bounded above by 2, and find its limit.

Exercise 9.6 MEDIUM

Using Theorem 9.12, find a closed form for $\sum_{k=1}^n (2k-1)$ (the sum of the first n odd numbers) and for $\sum_{k=1}^n k(k+1)$.

Exercise 9.7 MEDIUM

Determine the convergence of: (a) $\sum \frac{n}{2^n}$; (b) $\sum \frac{1}{\sqrt{n}}$; (c) $\sum \frac{(-1)^n}{\ln(n+1)}$.

Exercise 9.8 MEDIUM

Evaluate the telescoping series $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$.

Exercise 9.9 MEDIUM

Solve the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$, $a_0 = 1$, $a_1 = 0$.

Exercise 9.10 HARD

Solve $a_n = 4a_{n-1} - 4a_{n-2}$, $a_0 = 1$, $a_1 = 4$ (double root case).

Exercise 9.11 HARD

Prove by induction the bound $F_n \leq \varphi^{n-1}$ for $n \geq 1$, where $\varphi = \frac{1+\sqrt{5}}{2}$ (note $\varphi^2 = \varphi + 1$).

Exercise 9.12 HARD

Show that $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges although $\frac{1}{n \ln n} \rightarrow 0$, using the integral test.

Solutions to the Exercises

Solution 9.1. (a) $1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}$. (b) 2, 5, 14, 41, 122. ■

Solution 9.2. AP with $a_1 = 18$, $d = 3$, $n = 20$: last row has $18 + 19 \cdot 3 = 75$ seats; total $\frac{20}{2}(18 + 75) = 10 \cdot 93 = 930$. ■

Solution 9.3. $5 \frac{2^{10}-1}{2-1} = 5 \cdot 1023 = 5115$; $\frac{5}{1-1/3} = \frac{15}{2}$. ■

Solution 9.4. $\left| \frac{2n+1}{n+3} - 2 \right| = \frac{5}{n+3}$. Given $\varepsilon > 0$, choose $N > \frac{5}{\varepsilon} - 3$; then $n \geq N \Rightarrow \frac{5}{n+3} \leq \frac{5}{N+3} < \varepsilon$. ■

Solution 9.5. Bounded: by induction, $a_n < 2 \Rightarrow a_{n+1} = \sqrt{2+a_n} < \sqrt{4} = 2$, and $a_1 = \sqrt{2} < 2$. Increasing: $a_{n+1}^2 - a_n^2 = 2 + a_n - a_n^2 = (2 - a_n)(1 + a_n) > 0$. By monotone convergence the limit L exists and satisfies $L = \sqrt{2+L}$, i.e. $L^2 - L - 2 = 0$, so $L = 2$ (the root -1 being impossible). ■

Solution 9.6. $\sum_{k=1}^n (2k-1) = 2 \frac{n(n+1)}{2} - n = n^2$. $\sum k(k+1) = \sum k^2 + \sum k = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{3}$. ■

Solution 9.7. (a) Ratio test: $\frac{(n+1)/2^{n+1}}{n/2^n} = \frac{n+1}{2n} \rightarrow \frac{1}{2} < 1$: converges. (b) p -series with $p = \frac{1}{2} \leq 1$: diverges. (c) $\frac{1}{\ln(n+1)}$ decreases to 0: converges by Leibniz (only conditionally — compare with the divergent $\sum 1/(n \ln n)$ -type behaviour of the absolute series, which dominates $1/n$ eventually... in fact $\frac{1}{\ln(n+1)} > \frac{1}{n}$ for $n \geq 1$, so absolute convergence fails by comparison with the harmonic series). ■

Solution 9.8. $\frac{1}{(2n-1)(2n+1)} = \frac{1}{2} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right)$, so the partial sum is $\frac{1}{2} \left(1 - \frac{1}{2N+1} \right) \rightarrow \frac{1}{2}$. ■

Solution 9.9. Characteristic equation $x^2 = 5x - 6$: roots 2, 3. General solution $a_n = \alpha 2^n + \beta 3^n$; conditions give $\alpha + \beta = 1$, $2\alpha + 3\beta = 0$, so $\beta = -2$, $\alpha = 3$: $a_n = 3 \cdot 2^n - 2 \cdot 3^n$. ■

Solution 9.10. $x^2 = 4x - 4$ has the double root $r = 2$, so $a_n = (\alpha + \beta n)2^n$. From $a_0 = 1$: $\alpha = 1$; from $a_1 = 4$: $(1 + \beta) \cdot 2 = 4$, $\beta = 1$. Hence $a_n = (n + 1) 2^n$. ■

Solution 9.11. Strong induction. Base: $F_1 = 1 = \varphi^0$ and $F_2 = 1 \leq \varphi^1$. Step ($n \geq 3$): assuming the bound below n ,

$$F_n = F_{n-1} + F_{n-2} \leq \varphi^{n-2} + \varphi^{n-3} = \varphi^{n-3}(\varphi + 1) = \varphi^{n-3}\varphi^2 = \varphi^{n-1}.$$

■

Solution 9.12. $f(x) = \frac{1}{x \ln x}$ is positive and decreasing on $[2, \infty)$, and $\int_2^T \frac{dx}{x \ln x} = \ln \ln T - \ln \ln 2 \rightarrow \infty$. By the integral test the series diverges — terms tending to 0 is necessary but never sufficient (Theorem 9.14). ■

Part III

Counting

Permutations and Combinations

Notes By Pr. El Hadiq Zouhair

10.1 The basic counting rules

Theorem 10.1 (Sum rule). If a task can be done in n_1 ways or, *exclusively*, in n_2 ways, it can be done in $n_1 + n_2$ ways. Set version: disjoint sets satisfy $|A \cup B| = |A| + |B|$.

Theorem 10.2 (Product rule). If a procedure consists of two successive independent choices, with n_1 options for the first and n_2 for the second, there are $n_1 n_2$ outcomes. Set version: $|A \times B| = |A| |B|$.

Example 10.3. A password of 3 letters (26 choices each) followed by 2 digits: $26^3 \cdot 10^2 = 1,757,600$. A menu offering 4 starters or 6 mains as a single dish: $4 + 6 = 10$ choices. Tree diagrams display such combined choices as branches; the leaves count the outcomes.

10.2 Permutations

Definition 10.4. A *permutation* of n distinct objects is an ordering of all of them. The number of permutations is $n! = n(n-1)(n-2) \cdots 2 \cdot 1$, with $0! = 1$.

Theorem 10.5 (k -permutations). The number of ordered arrangements of k objects chosen from n distinct objects is

$$P(n, k) = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

Example 10.6 (Sprint podium). Nine athletes run a sprint; podium orders (gold, silver, bronze) number $P(9, 3) = 9 \cdot 8 \cdot 7 = 504$.

Theorem 10.7 (Permutations with repeated objects). The number of distinct arrangements of n objects of which n_1 are alike of type 1, $\dots$, n_k alike of type k ($n_1 + \dots + n_k = n$) is

$$\frac{n!}{n_1! n_2! \dots n_k!}.$$

Example 10.8. “MISSISSIPPI” has $\frac{11!}{1! 4! 4! 2!} = 34650$ distinct arrangements.

Theorem 10.9 (Circular permutations). n distinct people can be seated around a round table (rotations considered identical) in $(n - 1)!$ ways.

10.3 Combinations

Definition 10.10. A *combination* of k objects from n is an unordered selection. Their number is the *binomial coefficient*

$$C(n, k) = \binom{n}{k} = \frac{n!}{k! (n - k)!} = \frac{P(n, k)}{k!}.$$

Example 10.11. Choosing a committee of 5 from 30 students: $\binom{30}{5} = 142506$. Note order is irrelevant — choosing {Amal, Badr} equals choosing {Badr, Amal}.

Theorem 10.12 (Symmetry and Pascal’s identity).

$$\binom{n}{k} = \binom{n}{n - k}, \quad \binom{n + 1}{k} = \binom{n}{k - 1} + \binom{n}{k}.$$

Proof. Symmetry: choosing the k selected is the same as choosing the $n - k$ rejected. Pascal: fix one element x of an $(n + 1)$ -set; the k -subsets split into those containing x ($\binom{n}{k - 1}$ of them) and those not ($\binom{n}{k}$). $\square$

Pascal’s identity generates *Pascal’s triangle*, whose row n lists $\binom{n}{0}, \dots, \binom{n}{n}$:

$$\begin{array}{ccccccc} & & & & 1 & & & \\ & & & & 1 & & 1 & \\ & & & 1 & & 2 & & 1 \\ & & 1 & & 3 & & 3 & & 1 \\ 1 & & 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

Row sums give $\sum_k \binom{n}{k} = 2^n$ (each subset of an n -set has some size k — recounting the power-set theorem of Chapter 6).

10.4 The binomial theorem

Theorem 10.13 (Binomial theorem). For all x, y and $n \in \mathbb{N}$:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

Proof. Expanding $(x + y)(x + y) \cdots (x + y)$, each term arises by choosing y from k of the n factors and x from the rest; there are $\binom{n}{k}$ such choices, giving the coefficient of $x^{n-k} y^k$. $\square$

10.5 Combinations with repetition

Theorem 10.14 (Stars and bars). The number of multisets of size k from n types — equivalently the number of solutions of $x_1 + \cdots + x_n = k$ in nonnegative integers — is

$$\binom{n + k - 1}{k}.$$

Proof. Encode a selection as k stars separated by $n - 1$ bars; a word with $n + k - 1$ symbols is determined by the positions of the k stars. $\square$

10.5.1 The four counting problems

Select k from n	order matters	order irrelevant
without repetition	$P(n, k) = \frac{n!}{(n - k)!}$	$\binom{n}{k}$
with repetition	n^k	$\binom{n + k - 1}{k}$

10.6 Multinomial coefficients

Theorem 10.15 (Multinomial theorem).

$$(x_1 + \cdots + x_m)^n = \sum_{n_1 + \cdots + n_m = n} \binom{n}{n_1, n_2, \dots, n_m} x_1^{n_1} \cdots x_m^{n_m}, \quad \binom{n}{n_1, \dots, n_m} = \frac{n!}{n_1! \cdots n_m!},$$

the multinomial coefficient counting the ways to partition n labelled objects into groups of sizes $n_1, \dots, n_m$ (cf. Theorem 10.7).

10.7 The pigeonhole principle

Theorem 10.16 (Pigeonhole principle). If $n + 1$ or more objects are placed into n boxes, some box contains at least two objects. Generalised form: with N objects in n boxes, some box contains at least $\lceil N/n \rceil$ objects.

Example 10.17. Among any 13 people, two share a birth month. Among 100 people, at least $\lceil 100/12 \rceil = 9$ share one. Subtler: among any $n + 1$ integers chosen from $\{1, \dots, 2n\}$, two are consecutive — pair the numbers $\{1, 2\}, \{3, 4\}, \dots, \{2n-1, 2n\}$ and apply pigeonhole.

10.8 Inclusion–exclusion

Theorem 10.18. For finite sets $A_1, \dots, A_n$:

$$\left| \bigcup_i A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap \dots \cap A_n|.$$

Example 10.19 (Derangements). A *derangement* is a permutation with no fixed point. Let A_i be the permutations fixing i ; inclusion–exclusion gives

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \xrightarrow{n \rightarrow \infty} \frac{n!}{e}.$$

So if n guests grab random hats, the chance nobody gets their own tends to $1/e \approx 0.3679$.

10.9 Two classical identities

Theorem 10.20 (Vandermonde). $\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$ — choose r people from m women and n men by cases on how many women.

Theorem 10.21 (Hockey stick). $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$ — classify the $(r+1)$ -subsets of $\{0, \dots, n\}$ by their largest element.

Example 10.22 (Poker hands). From a 52-card deck, $\binom{52}{5} = 2598960$ five-card hands. Full houses: choose the triple rank, its suits, the pair rank, its suits: $13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2} = 3744$.

10.10 Strategy: choosing the right formula

Ask in order: (1) Am I *selecting* or *arranging* (does order matter)? (2) Is repetition allowed? — this places you in the four-problem table. (3) Are objects distinguishable? If types repeat, use multiset permutations or stars and bars. (4) “At least/at most” constraints often yield to complement counting or inclusion–exclusion; existence claims to pigeonhole.

10.11 Summary

Sum and product rules compose all elementary counts. Order + no repetition: $P(n, k)$; no order: $\binom{n}{k}$; with repetition: n^k and $\binom{n+k-1}{k}$. Pascal's identity builds the triangle; the binomial and multinomial theorems expand powers; pigeonhole guarantees collisions; inclusion–exclusion corrects overcounting — derangements being its signature application.

Exercises

Exercise 10.1 EASY

License plates consist of 2 letters followed by 3 digits. How many are possible? How many with no repeated symbol?

Exercise 10.2 EASY

Compute $P(7, 3)$, $\binom{7}{3}$, and the number of binary strings of length 7 containing exactly three 1s.

Exercise 10.3 EASY

How many distinct arrangements has the word “ABRACADABRA”?

Exercise 10.4 MEDIUM

A committee of 6 is chosen from 10 men and 8 women. How many committees contain (a) exactly 3 women; (b) at least one woman?

Exercise 10.5 MEDIUM

How many solutions in nonnegative integers has $x + y + z = 11$? How many with $x \geq 2$?

Exercise 10.6 MEDIUM

Find the coefficient of x^5 in $(2x - 3)^8$.

Exercise 10.7 MEDIUM

In how many ways can 8 people sit around a round table? What if two specific people insist on sitting together?

Exercise 10.8 MEDIUM

Show that among any $n + 1$ integers from $\{1, 2, \dots, 2n\}$, one divides another. *Hint: write each as $2^a m$ with m odd, and apply pigeonhole to the odd parts.*

Exercise 10.9 MEDIUM

How many integers in $\{1, \dots, 1000\}$ are divisible by 2, 3 or 5?

Exercise 10.10 HARD

Prove Vandermonde's identity (Theorem 10.20) by computing the coefficient of x^r in $(1+x)^m(1+x)^n = (1+x)^{m+n}$.

Exercise 10.11 HARD

Four guests check four hats and receive them back at random. Using Example 10.19, count the outcomes where nobody receives their own hat, and the probability of that event.

Exercise 10.12 HARD

How many surjections are there from a 5-element set onto a 3-element set? *Hint: inclusion-exclusion over the missed elements: $\sum_j (-1)^j \binom{3}{j} (3-j)^5$.*

Solutions to the Exercises

Solution 10.1. $26^2 \cdot 10^3 = 676000$. Without repetition: $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8 = 468000$. ■

Solution 10.2. $P(7, 3) = 7 \cdot 6 \cdot 5 = 210$; $\binom{7}{3} = 35$; a string is determined by the positions of its three 1s: $\binom{7}{3} = 35$. ■

Solution 10.3. Letters: A $\times 5$, B $\times 2$, R $\times 2$, C, D — total 11: $\frac{11!}{5!2!2!1!1!} = 83160$. ■

Solution 10.4. (a) $\binom{8}{3} \binom{10}{3} = 56 \cdot 120 = 6720$. (b) Total minus all-male: $\binom{18}{6} - \binom{10}{6} = 18564 - 210 = 18354$. ■

Solution 10.5. Stars and bars ($n = 3$, $k = 11$): $\binom{13}{2} = 78$. With $x \geq 2$, substitute $x' = x - 2$: $x' + y + z = 9$, giving $\binom{11}{2} = 55$. ■

Solution 10.6. General term $\binom{8}{k} (2x)^{8-k} (-3)^k$; x^5 needs $k = 3$: $\binom{8}{3} 2^5 (-3)^3 = 56 \cdot 32 \cdot (-27) = -48384$. ■

Solution 10.7. $(8 - 1)! = 5040$. Glue the pair into one block: $(7 - 1)! = 720$ circular orders, times 2 internal arrangements: 1440. ■

Solution 10.8. Write each chosen integer as $2^a m$ with odd part $m \in \{1, 3, \dots, 2n - 1\}$ — only n possible odd parts for $n + 1$ numbers, so two share the same m : say $2^a m$ and $2^b m$ with $a < b$. Then $2^a m \mid 2^b m$. ■

Solution 10.9. With $D_k =$ multiples of k : $|D_2| = 500$, $|D_3| = 333$, $|D_5| = 200$, $|D_6| = 166$, $|D_{10}| = 100$, $|D_{15}| = 66$, $|D_{30}| = 33$. Inclusion-exclusion: $500 + 333 + 200 - 166 - 100 - 66 + 33 = 734$. ■

Solution 10.10. The coefficient of x^r in $(1 + x)^{m+n}$ is $\binom{m+n}{r}$. In the product $(1 + x)^m (1 + x)^n = (\sum_k \binom{m}{k} x^k) (\sum_j \binom{n}{j} x^j)$, the x^r terms pair $k + j = r$: coefficient $\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$. Equate. ■

Solution 10.11. $D_4 = 4! (1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}) = 24 \cdot \frac{9}{24} = 9$ derangements out of $4! = 24$ outcomes: probability $\frac{9}{24} = \frac{3}{8} = 0.375$ (already close to $1/e$). ■

Solution 10.12. $\sum_{j=0}^3 (-1)^j \binom{3}{j} (3-j)^5 = 3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 - 0 = 243 - 96 + 3 = 150$. ■

Binomial Identities

Notes By Pr. El Hadiq Zouhair

11.1 Two kinds of proof

Recall $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts the k -subsets of an n -set. Binomial identities admit two complementary proof styles: *algebraic* (manipulate factorials, or compare coefficients in the binomial theorem) and *combinatorial* (“double counting”: count one set in two ways). Whenever possible we give both — the combinatorial proof usually explains *why* the identity holds.

11.2 The fundamental identities

Theorem 11.1 (Symmetry). $\binom{n}{k} = \binom{n}{n-k}$.

Proof. Algebraic: both equal $\frac{n!}{k!(n-k)!}$. Combinatorial: selecting k elements is the same as rejecting $n-k$. $\square$

Theorem 11.2 (Pascal). $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$.

Proof. Fix a distinguished element x of an $(n+1)$ -set: k -subsets containing x number $\binom{n}{k-1}$, those avoiding x number $\binom{n}{k}$. $\square$

Pascal’s triangle hides many patterns: the hockey-stick sums below, the Fibonacci numbers along shallow diagonals ($\sum_k \binom{n-k}{k} = F_{n+1}$), the parity fractal of Sierpiński, and the central column $\binom{2n}{n}$ governing Catalan numbers.

Theorem 11.3 (Absorption). $k \binom{n}{k} = n \binom{n-1}{k-1}$, equivalently $\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$.

Proof. Count pairs (committee of k , its chair) from n people in two ways: choose the committee then the chair $\left(\binom{n}{k} \cdot k\right)$, or the chair then the rest $\left(n \cdot \binom{n-1}{k-1}\right)$. $\square$

11.3 The binomial theorem and its harvest

Theorem 11.4 (Binomial theorem). $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$.

Substituting values of x, y yields identities mechanically.

Theorem 11.5 (Row sum). $\sum_{k=0}^n \binom{n}{k} = 2^n$.

Proof. Three proofs. (1) Set $x = y = 1$ in the binomial theorem. (2) Combinatorial: both sides count subsets of an n -set, the left side by size. (3) Induction via Pascal's identity. $\square$

Theorem 11.6 (Alternating sum). $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$ for $n \geq 1$ — set $x = 1, y = -1$. Hence even-sized subsets equal odd-sized subsets in number (2^{n-1} each).

Theorem 11.7 (Weighted sums).

$$\sum_{k=0}^n 2^k \binom{n}{k} = 3^n, \quad \sum_{k=0}^n k \binom{n}{k} = n 2^{n-1}, \quad \sum_{k=0}^n k^2 \binom{n}{k} = n(n+1) 2^{n-2}.$$

Proof. First: $x = 1, y = 2$. Second: absorption gives $\sum_k k \binom{n}{k} = n \sum_k \binom{n-1}{k-1} = n 2^{n-1}$; combinatorially, count (subset, distinguished member) pairs. Third: write $k^2 = k(k-1) + k$ and apply absorption twice: $\sum k(k-1) \binom{n}{k} = n(n-1) 2^{n-2}$, then add $n 2^{n-1}$ to get $n(n-1) 2^{n-2} + n 2^{n-1} = n(n+1) 2^{n-2}$. $\square$

11.4 Vandermonde and friends

Theorem 11.8 (Vandermonde). $\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$.

Example 11.9. A jury of 4 from 6 magistrates and 5 citizens: $\binom{11}{4} = 330$ directly; by cases on magistrates: $\binom{6}{0} \binom{5}{4} + \binom{6}{1} \binom{5}{3} + \binom{6}{2} \binom{5}{2} + \binom{6}{3} \binom{5}{1} + \binom{6}{4} \binom{5}{0} = 5 + 60 + 150 + 100 + 15 = 330$. $\checkmark$

Theorem 11.10 (Sum of squares). $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$ — Vandermonde with $m = n = r$, using symmetry $\binom{n}{n-k} = \binom{n}{k}$.

Theorem 11.11 (Hockey stick). $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$.

Proof. Classify $(r+1)$ -subsets of $\{1, \dots, n+1\}$ by largest element $i+1$: the remaining r elements come from $\{1, \dots, i\}$, in $\binom{i}{r}$ ways, for $i = r, \dots, n$. $\square$

11.5 Extensions

Definition 11.12 (Generalised binomial coefficient). For real α and $k \in \mathbb{N}$: $\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!}$. Newton's binomial series $(1+x)^\alpha = \sum_k \binom{\alpha}{k} x^k$ converges for $|x| < 1$; the *negative-binomial* case gives $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$, linking to stars and bars.

Definition 11.13 (Catalan numbers). $C_n = \frac{1}{n+1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}$: 1, 1, 2, 5, 14, 42, ... They count balanced parenthesisations, monotonic lattice paths not crossing the diagonal, triangulations of convex polygons, and binary trees.

The multinomial coefficients $\binom{n}{n_1, \dots, n_m}$ extend the whole theory from two blocks to m ; and *generating functions* — viewing $\sum_k \binom{n}{k} x^k = (1+x)^n$ as a polynomial identity — turn combinatorial convolutions into products of series, the method behind the Vandermonde proof and far beyond.

11.6 The combinatorial-proof template

To prove LHS = RHS combinatorially: (1) pose a concrete counting question; (2) show the LHS answers it one way; (3) show the RHS answers the same question another way; (4) conclude the counts are equal. Practise the template on the worked example below.

Example 11.14 (Portfolio choice). Maeve must pick a team of k analysts from n candidates and name one team lead, but she may also count by first naming the lead (n ways) and then completing the team ($\binom{n-1}{k-1}$ ways). The two counts, $k \binom{n}{k}$ and $n \binom{n-1}{k-1}$, must agree: absorption, proved by the template.

11.7 Master cheat sheet

Symmetry	$\binom{n}{k} = \binom{n}{n-k}$
Pascal	$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$
Absorption	$k\binom{n}{k} = n\binom{n-1}{k-1}$
Row sum	$\sum_k \binom{n}{k} = 2^n$
Alternating	$\sum_k (-1)^k \binom{n}{k} = 0 \quad (n \geq 1)$
Weighted	$\sum_k k\binom{n}{k} = n2^{n-1}, \quad \sum_k k^2\binom{n}{k} = n(n+1)2^{n-2}$
Powers of 3	$\sum_k 2^k \binom{n}{k} = 3^n$
Vandermonde	$\binom{m+n}{r} = \sum_k \binom{m}{k} \binom{n}{r-k}$
Squares	$\sum_k \binom{n}{k}^2 = \binom{2n}{n}$
Hockey stick	$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$
Catalan	$C_n = \frac{1}{n+1} \binom{2n}{n}$

Common pitfalls: forgetting that $\binom{n}{k} = 0$ for $k < 0$ or $k > n$; shifting summation indices incorrectly; applying symmetry inside a sum without adjusting the range; and assuming an identity for real upper index when it only holds for integers.

11.8 Summary

Every identity in this chapter flows from three sources: Pascal's recurrence, the binomial theorem with clever substitutions, and double counting. Master the cheat sheet, but more importantly master the *template*: a binomial identity is an invitation to find the single question that both sides answer.

Exercises

Exercise 11.1 EASY

Evaluate without a calculator: $\binom{10}{8}, \binom{12}{1}, \binom{9}{4} + \binom{9}{5}$.

Exercise 11.2 EASY

Use the row-sum and alternating-sum identities to find $\sum_{k \text{ even}} \binom{n}{k}$ for $n \geq 1$.

Exercise 11.3 EASY

Verify the hockey-stick identity $\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \binom{5}{2} = \binom{6}{3}$ numerically.

Exercise 11.4 MEDIUM

Give a combinatorial proof of $\binom{n}{2} \binom{n-2}{k-2} = \binom{n}{k} \binom{k}{2}$ ($2 \leq k \leq n$).

Exercise 11.5 MEDIUM

Evaluate $\sum_{k=0}^n 3^k \binom{n}{k}$ and $\sum_{k=0}^n (-2)^k \binom{n}{k}$.

Exercise 11.6 MEDIUM

Prove $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$ combinatorially, by counting committees with a chair.

Exercise 11.7 MEDIUM

Using Vandermonde, evaluate $\sum_{k=0}^m \binom{m}{k} \binom{n}{p+k}$ as a single binomial coefficient ($p+m \leq n$).

Exercise 11.8 MEDIUM

Show that $\binom{2n}{n}$ is even for $n \geq 1$. *Hint:* $\binom{2n}{n} = \binom{2n-1}{n-1} + \binom{2n-1}{n}$ and symmetry.

Exercise 11.9 HARD

Prove the Fibonacci diagonal identity $\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} = F_{n+1}$ (with $F_1 = F_2 = 1$), by induction using Pascal's identity.

Exercise 11.10 HARD

Prove $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$ combinatorially, by counting n -subsets of a set of n women and n men.

Exercise 11.11 HARD

Show $C_n = \binom{2n}{n} - \binom{2n}{n+1}$ counts lattice paths from $(0, 0)$ to (n, n) (unit right/up steps) that never rise above the diagonal, via the reflection principle.

Solutions to the Exercises

Solution 11.1. $\binom{10}{8} = \binom{10}{2} = 45$; $\binom{12}{1} = 12$; $\binom{9}{4} + \binom{9}{5} = \binom{10}{5} = 252$ (Pascal). ■

Solution 11.2. Adding $\sum_k \binom{n}{k} = 2^n$ and $\sum_k (-1)^k \binom{n}{k} = 0$ cancels odd k and doubles even k : $\sum_{k \text{ even}} \binom{n}{k} = 2^{n-1}$. ■

Solution 11.3. $1 + 3 + 6 + 10 = 20 = \binom{6}{3}$. ✓ ■

Solution 11.4. Both sides count (committee of k from n people, with 2 designated co-chairs): left side picks the co-chairs first ($\binom{n}{2}$), then fills the committee ($\binom{n-2}{k-2}$); right side picks the committee ($\binom{n}{k}$), then its co-chairs ($\binom{k}{2}$). ■

Solution 11.5. Binomial theorem with $(x, y) = (1, 3)$: 4^n . With $(1, -2)$: $(-1)^n$. ■

Solution 11.6. Count pairs (committee S of any size, chair $c \in S$) from n people. By size: choose size k , the committee, then the chair: $\sum_k \binom{n}{k} k$. Chair first: n choices, then any subset of the remaining $n-1$ people joins: $n2^{n-1}$. Equate. ■

Solution 11.7. By symmetry $\binom{m}{k} = \binom{m}{m-k}$, the sum is $\sum_k \binom{m}{m-k} \binom{n}{p+k}$, a Vandermonde convolution picking $m+p$ objects from $m+n$: $\binom{m+n}{m+p}$. ■

Solution 11.8. Pascal: $\binom{2n}{n} = \binom{2n-1}{n-1} + \binom{2n-1}{n}$, and the two terms are equal by symmetry $((n-1) + n = 2n-1)$. Hence $\binom{2n}{n} = 2\binom{2n-1}{n-1}$ is even. ■

Solution 11.9. Let $S_n = \sum_k \binom{n-k}{k}$. Check $S_1 = 1 = F_2$, $S_2 = 2 = F_3$. For the step, Pascal gives $\binom{n-k}{k} = \binom{n-1-k}{k} + \binom{n-1-k}{k-1}$; summing over k , the first parts sum to S_{n-1} and the second parts re-index $(k \rightarrow k-1)$ to S_{n-2} . So $S_n = S_{n-1} + S_{n-2}$ — the Fibonacci recurrence with Fibonacci initial values, hence $S_n = F_{n+1}$ for all n . ■

Solution 11.10. Choosing n people from n women + n men splits by the number k of women chosen: $\binom{n}{k}$ ways for the women and $\binom{n}{n-k} = \binom{n}{k}$ for the men, giving $\sum_k \binom{n}{k}^2$. Direct count: $\binom{2n}{n}$. Equate. ■

Solution 11.11. All paths to (n, n) : $\binom{2n}{n}$. A “bad” path touches the line $y = x + 1$; reflecting its prefix up to the first touch across that line bijects bad paths with paths from $(-1, 1)$ to (n, n) , which number $\binom{2n}{n+1}$. Good paths: $\binom{2n}{n} - \binom{2n}{n+1} = C_n$. ■

Part IV

Proofs and Recursion

Proof Techniques

Notes By Pr. El Hadiq Zouhair

12.1 The vocabulary of mathematical statements

A *theorem* is a statement proved true; a *lemma* is a helper theorem; a *corollary* follows easily from a theorem; a *conjecture* awaits proof; a *proof* is a valid argument, built from axioms, definitions and previous results by the rules of inference of Chapter 5. Most theorems have the form $\forall x (P(x) \rightarrow Q(x))$, and the techniques below are strategies for establishing $P \rightarrow Q$ for an arbitrary x (whence UG concludes).

12.2 Direct proof

To prove $p \rightarrow q$ directly: assume p , derive q by definitions and known results.

Theorem 12.1. If n is odd then n^2 is odd.

Proof. Let $n = 2k + 1$ for some integer k . Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. $\square$

12.3 Proof by contraposition

To prove $p \rightarrow q$, prove $\neg q \rightarrow \neg p$ instead — the two are equivalent (Chapter 2).

Theorem 12.2. If n^2 is even then n is even.

Proof. Contrapositive: if n is odd then n^2 is odd — proved above. $\square$

Choose contraposition when $\neg q$ is more concrete to work with than p (here: “ n odd” gives an explicit form, “ n^2 even” does not).

12.4 Proof by contradiction

To prove φ : assume $\neg\varphi$ and derive a contradiction ($r \wedge \neg r$); then $\neg\varphi$ is impossible, so φ holds.
Template: “Suppose, for contradiction, that ... Then ... contradiction. Hence ...”

Theorem 12.3 (Canonical example). $\sqrt{2}$ is irrational.

Proof. Suppose $\sqrt{2} = a/b$ with $a, b \in \mathbb{Z}^+$ in lowest terms. Then $a^2 = 2b^2$, so a^2 is even, so a is even (previous theorem): $a = 2c$. Then $4c^2 = 2b^2$, i.e. $b^2 = 2c^2$, so b is even too — contradicting lowest terms. Hence $\sqrt{2}$ is irrational. $\square$

Theorem 12.4 (Euclid). There are infinitely many primes.

Proof. Suppose $p_1, \dots, p_k$ were all the primes and set $N = p_1 \cdots p_k + 1$. Each p_i divides $N - 1$, so none divides N ; yet $N > 1$ has a prime divisor — a prime outside the list: contradiction. $\square$

12.5 Proof by cases

To prove φ , split the situation into finitely many exhaustive cases and prove φ in each — justified by $(p_1 \vee \cdots \vee p_n) \wedge \bigwedge_i (p_i \rightarrow q) \models q$.

Theorem 12.5. For every integer n , $n^2 \geq n$.

Proof. Case $n \leq 0$: then $n^2 \geq 0 \geq n$. Case $n \geq 1$: multiplying $n \geq 1$ by $n > 0$ gives $n^2 \geq n$. The cases are exhaustive. $\square$

12.6 Existence proofs

An existence statement $\exists x P(x)$ may be proved *constructively* — exhibit a witness — or *non-constructively* — show a witness must exist without naming one.

Example 12.6 (Constructive). There is a positive integer expressible as a sum of two cubes in two ways: $1729 = 1^3 + 12^3 = 9^3 + 10^3$.

Example 12.7 (Non-constructive). There exist irrationals x, y with x^y rational. Consider $z = \sqrt{2}^{\sqrt{2}}$. If z is rational, take $x = y = \sqrt{2}$. If z is irrational, take $x = z, y = \sqrt{2}$: then $x^y = \sqrt{2}^2 = 2$. Either way a witness exists — yet we have not determined which.

12.7 Uniqueness proofs

To prove $\exists! x P(x)$: prove existence, then prove uniqueness — assume $P(a)$ and $P(b)$ and derive $a = b$.

Example 12.8. The equation $ax = b$ ($a \neq 0$) has the unique solution $x = b/a$: existence by substitution; uniqueness since $ax_1 = b = ax_2$ implies $x_1 = x_2$ after dividing by $a \neq 0$.

12.8 Disproof by counterexample

To refute $\forall x P(x)$, one counterexample suffices.

Example 12.9. “Every positive integer is a sum of two squares” is false: 3 is not. “ $2^{2^n} + 1$ is prime for all n ” (Fermat) fails at $n = 5$: $2^{32} + 1 = 641 \times 6700417$ (Euler).

12.9 Common errors

Beware: proving $p \rightarrow q$ by assuming q (begging the question); arguing from examples (“it works for $n = 1, 2, 3$, hence always”); dividing by a quantity that may be zero; affirming the consequent; and the hidden assumption that objects named differently are distinct. A classic bogus proof that $1 = 2$ divides both sides of $a^2 - ab = a^2 - b^2$ by $a - b$ when $a = b$ — division by zero in disguise.

12.10 Strategy

Given a conjecture $p \rightarrow q$: try *direct* first; if $\neg q$ unpacks better, try *contraposition*; if both directions stall or the statement is a pure negative (“is irrational”, “no largest ...”), try *contradiction*; if natural subcases appear (parity, sign, remainders), use *cases*. For $\exists$, look for an explicit witness before resorting to non-constructive arguments. And before proving anything hard, hunt briefly for a counterexample — failure to find one often reveals the proof idea. Statements quantified over all $n \in \mathbb{N}$ usually call for *induction*, the subject of Chapter 13.

12.11 Summary

Direct, contrapositive, contradiction and cases prove implications; witnesses or indirect arguments prove existence; existence plus “two solutions are equal” proves uniqueness; one counterexample refutes a universal claim. The two jewels — irrationality of $\sqrt{2}$ (Theorem 12.3) and Euclid’s infinitude of primes (Theorem 12.4) — model the contradiction template every mathematician carries.

Exercises

Exercise 12.1 EASY

Prove directly: the sum of two odd integers is even.

Exercise 12.2 EASY

Prove directly: if $a \mid b$ and $b \mid c$ then $a \mid c$.

Exercise 12.3 EASY

Find counterexamples: (a) “every prime is odd”; (b) “if $n^2 > 0$ then $n > 0$ ”; (c) “the sum of two irrationals is irrational”.

Exercise 12.4 MEDIUM

Prove by contraposition: if $3n + 2$ is odd, then n is odd.

Exercise 12.5 MEDIUM

Prove by contradiction: there is no smallest positive rational number.

Exercise 12.6 MEDIUM

Prove by cases: for all integers n , $3 \mid n(n+1)(n+2)$.

Exercise 12.7 MEDIUM

Prove that $\sqrt{3}$ is irrational, adapting the proof of Theorem 12.3. Where would the same argument break for $\sqrt{4}$?

Exercise 12.8 MEDIUM

Prove there exists a unique real x with $x^3 + x - 2 = 0$. (*Existence: exhibit it. Uniqueness: factor or use monotonicity.*)

Exercise 12.9 HARD

Prove that if n is a positive integer and n^2 is divisible by 3, then n is divisible by 3 (use cases on $n \bmod 3$); this is the lemma your $\sqrt{3}$ proof needed.

Exercise 12.10 HARD

Show non-constructively that in any group of 2 or more people at a party, two people have shaken the same number of hands (handshakes within the group). *Hint: pigeonhole on the possible counts.*

Exercise 12.11 HARD

Criticise this “proof” that every positive integer n equals the next: “ $n = n + 1$ implies $(n)^2 - (n + 1)n = n^2 - (n + 1)n$, both sides simplify to $-n = -n$, true; hence $n = n + 1$.” Identify the logical error precisely.

Solutions to the Exercises

Solution 12.1. Let $m = 2j + 1$, $n = 2k + 1$. Then $m + n = 2(j + k + 1)$, an even number. ■

Solution 12.2. $b = aj$ and $c = bk$ for integers j, k ; then $c = a(jk)$, so $a \mid c$. ■

Solution 12.3. (a) 2. (b) $n = -1$: $n^2 = 1 > 0$ but $n < 0$. (c) $\sqrt{2} + (-\sqrt{2}) = 0 \in \mathbb{Q}$. ■

Solution 12.4. Contrapositive: if n is even, $3n+2$ is even. Let $n = 2k$: $3n+2 = 6k+2 = 2(3k+1)$, even. Done. ■

Solution 12.5. Suppose $r > 0$ were the smallest positive rational. Then $r/2$ is rational, positive, and smaller — contradiction. ■

Solution 12.6. Among any three consecutive integers one is divisible by 3: by cases on $n \bmod 3$. If $n \equiv 0$, then $3 \mid n$; if $n \equiv 1$, then $3 \mid n + 2$; if $n \equiv 2$, then $3 \mid n + 1$. In each case the product is divisible by 3. ■

Solution 12.7. Suppose $\sqrt{3} = a/b$ in lowest terms; then $a^2 = 3b^2$, so $3 \mid a^2$, hence $3 \mid a$ (Exercise 12.9): $a = 3c$, so $3b^2 = 9c^2$, $b^2 = 3c^2$, and $3 \mid b$ — contradicting lowest terms. For $\sqrt{4}$, the lemma fails: $4 \mid a^2$ does *not* force $4 \mid a$ (take $a = 2$), and indeed $\sqrt{4} = 2$ is rational. ■

Solution 12.8. Existence: $x = 1$ works: $1 + 1 - 2 = 0$. Uniqueness: $x^3 + x - 2 = (x - 1)(x^2 + x + 2)$, and $x^2 + x + 2 = (x + \frac{1}{2})^2 + \frac{7}{4} > 0$ has no real roots; alternatively $f(x) = x^3 + x - 2$ is strictly increasing ($f'(x) = 3x^2 + 1 > 0$), so it crosses zero exactly once. ■

Solution 12.9. Cases. If $n = 3k$: done, $3 \mid n$. If $n = 3k + 1$: $n^2 = 9k^2 + 6k + 1 \equiv 1 \pmod{3}$, not divisible. If $n = 3k + 2$: $n^2 = 9k^2 + 12k + 4 \equiv 1 \pmod{3}$, not divisible. So $3 \mid n^2$ occurs only in the first case, where $3 \mid n$. ■

Solution 12.10. With n people, each handshake count lies in $\{0, 1, \dots, n - 1\}$ — n values. But counts 0 and $n - 1$ cannot both occur (someone who shook everyone's hand rules out someone who shook none). So at most $n - 1$ values are available for n people: by pigeonhole two people share a count. No specific pair is exhibited — the proof is non-constructive. ■

Solution 12.11. The argument assumes $n = n + 1$ *first* and derives a true statement. Deriving a truth from an assumption proves nothing about the assumption (that is affirming the consequent at the level of proofs); a valid proof must derive the *conclusion* from accepted premises, not the reverse. The implication chain is also reversible only if every step is an equivalence, which squaring-style manipulations are not. ■

Mathematical Induction

Notes By Pr. El Hadiq Zouhair

13.1 The well-ordering principle

Theorem 13.1 (Well-ordering principle, WOP). Every nonempty subset of $\mathbb{N}$ has a least element.

This innocuous statement is the bedrock: it is logically equivalent to the principle of induction, and powers “minimal counterexample” arguments — if a statement failed for some n , there would be a *smallest* failing n , which one then contradicts.

13.2 Weak induction

Theorem 13.2 (Principle of mathematical induction, PWI). Let $P(n)$ be a predicate on $\mathbb{N}$. If

- (i) **Base case:** $P(0)$ holds (or $P(n_0)$ for a starting point n_0), and
 - (ii) **Inductive step:** for all $n \geq n_0$, $P(n) \Rightarrow P(n+1)$,
- then $P(n)$ holds for all $n \geq n_0$.

WOP implies PWI. Suppose (i) and (ii) hold but P fails somewhere. By WOP let m be the least failure. $m \neq n_0$ by (i), so $m-1 \geq n_0$ and $P(m-1)$ holds (minimality). By (ii), $P(m)$ holds — contradiction. (Conversely PWI implies WOP: see Exercise 13.10.) $\square$

The template. To prove $\forall n \geq n_0, P(n)$: (1) *state* $P(n)$ clearly; (2) verify $P(n_0)$; (3) fix $n \geq n_0$, assume the *inductive hypothesis* $P(n)$, and derive $P(n+1)$ — pointing precisely at where the hypothesis is used; (4) conclude by induction.

13.3 Summation identities

Example 13.3. $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ for $n \geq 1$. *Base:* $1 = \frac{1 \cdot 2}{2}$. *Step:* assuming the identity for n ,

$$\sum_{i=1}^{n+1} i = \frac{n(n+1)}{2} + (n+1) = (n+1) \left(\frac{n}{2} + 1 \right) = \frac{(n+1)(n+2)}{2}. \quad \square$$

Example 13.4. $\sum_{k=0}^n 2^k = 2^{n+1} - 1$: base $1 = 2 - 1$; step: add 2^{n+1} to both sides: $2^{n+1} - 1 + 2^{n+1} = 2^{n+2} - 1$. Similarly $\sum_{k=1}^n k \cdot k! = (n+1)! - 1$: step uses $(n+1)!(n+2) - 1 - \dots$ — see Exercise 13.2. The sums $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$ and $\sum k^3 = \left(\frac{n(n+1)}{2} \right)^2$ follow the identical pattern (Chapter 9).

13.4 Inequalities

Example 13.5. $2^n > n^2$ for all $n \geq 5$. *Base:* $2^5 = 32 > 25$. *Step:* for $n \geq 5$, assume $2^n > n^2$. Then

$$2^{n+1} = 2 \cdot 2^n > 2n^2 = n^2 + n \cdot n \geq n^2 + 5n > n^2 + 2n + 1 = (n+1)^2,$$

using $n \geq 5 \Rightarrow 5n > 2n + 1$. Note the base case 5 is essential: the claim is false for $n = 2, 3, 4$.

Example 13.6. $n! \geq 2^n$ for $n \geq 4$: base $24 \geq 16$; step: $(n+1)! = (n+1) \cdot n! \geq (n+1)2^n \geq 2 \cdot 2^n = 2^{n+1}$ since $n+1 \geq 2$. Two classical companions provable the same way: Bernoulli's inequality $(1+x)^n \geq 1+nx$ for $x \geq -1$, and $\ln(n+1) \leq H_n = \sum_{i=1}^n \frac{1}{i}$.

13.5 Divisibility

Example 13.7. $3 \mid (n^3 + 2n)$ for all $n \geq 0$. *Base:* $3 \mid 0$. *Step:*

$$(n+1)^3 + 2(n+1) = n^3 + 3n^2 + 3n + 1 + 2n + 2 = (n^3 + 2n) + 3(n^2 + n + 1),$$

a sum of two multiples of 3 (the first by hypothesis).

13.6 Sets and geometry

Example 13.8 (Subsets). A set with n elements has 2^n subsets. *Base:* $\mathcal{P}(\emptyset) = \{\emptyset\}$, of size $1 = 2^0$. *Step:* let $|A| = n+1$, pick $x \in A$, $A' = A \setminus \{x\}$. Subsets of A split into subsets of A' and those plus x : $2^n + 2^n = 2^{n+1}$ by hypothesis.

Example 13.9 (Triangulation). Every convex polygon with $n \geq 3$ vertices can be divided into $n-2$ triangles by non-crossing diagonals. *Base:* a triangle, $1 = 3-2$. *Step:* in a convex

$(n+1)$ -gon, the diagonal from v_1 to v_n cuts off triangle $v_1v_nv_{n+1}$ and leaves a convex n -gon, triangulated by hypothesis into $n-2$ triangles: total $n-1 = (n+1) - 2$.

13.7 Strong induction

Theorem 13.10 (Strong induction). If $P(n_0)$ holds and, for all $n \geq n_0$,

$$(P(n_0) \wedge P(n_0+1) \wedge \cdots \wedge P(n)) \Rightarrow P(n+1),$$

then $P(n)$ holds for all $n \geq n_0$. Strong and weak induction are equivalent in power, but strong induction is the natural tool when $P(n+1)$ depends on values *further back* than n .

Example 13.11 (Prime factorisation). Every integer $n \geq 2$ is a product of primes. If $n+1$ is prime, done. Otherwise $n+1 = ab$ with $2 \leq a, b \leq n$; by the strong hypothesis both a and b are products of primes, hence so is $n+1$. (Weak induction is helpless here: a, b are far below $n+1$.)

Example 13.12 (Fibonacci bound). $F_n < 2^n$ for all $n \geq 0$: bases $F_0 = 0 < 1$, $F_1 = 1 < 2$; step ($n \geq 1$): $F_{n+1} = F_n + F_{n-1} < 2^n + 2^{n-1} < 2^{n+1}$, using the hypothesis at *two* previous points.

Example 13.13 (Postage stamps). Every amount $n \geq 12$ can be paid with 4- and 5-dirham stamps. Bases: $12 = 4 + 4 + 4$, $13 = 4 + 4 + 5$, $14 = 4 + 5 + 5$, $15 = 5 + 5 + 5$. Step for $n+1 \geq 16$: by strong hypothesis $(n+1) - 4 \geq 12$ is payable; add one 4-stamp.

13.8 Structural induction

For recursively defined structures (Chapter 14) — formulas, strings, trees — induction follows the structure: prove the property for the base objects, and show each construction rule preserves it. Example: every wff of propositional logic has as many left as right parentheses — true for atoms (none), and each formation rule adds one of each.

13.9 Pitfalls

Example 13.14 (All horses are the same colour). “Proof” that any n horses share a colour: for the step, remove one horse — the remaining n share a colour by hypothesis; remove a different one — same conclusion; the overlap forces all $n+1$ to match. The argument *fails* at $n+1 = 2$: the two groups of one horse do not overlap, so nothing links them. One broken link $(P(1) \rightarrow P(2))$ destroys the whole chain.

Always check: (1) the base case is actually verified (a missing or false base “proves” $\sum_{i=1}^n i = \frac{n(n+1)}{2} + 1$ too, since the step goes through!); (2) the inductive step is valid for every n from the base upward, including the very first; (3) the inductive hypothesis is genuinely used — otherwise the proof is direct, or wrong.

13.10 Induction and programs

Induction is the mathematics of loops and recursion: proving that a recursive program is correct *is* an induction on the input (see Chapter 14), and a loop is proved correct via a *loop invariant* — a property maintained by every iteration, i.e. an inductive step. Example: in the iterative factorial, the invariant “after k iterations the accumulator equals $k!$ ” is established at $k = 0$ and preserved each pass, so on exit (at $k = n$) the result is $n!$.

13.11 Summary

WOP, weak induction and strong induction are equivalent foundations. The template — state $P(n)$, verify the base, derive $P(n + 1)$ using the hypothesis visibly — proves summation identities, inequalities (mind the true base case), divisibility facts, set-theoretic and geometric claims. Strong induction handles recurrences reaching far back (prime factorisation, Fibonacci); structural induction handles recursively built objects; and loop invariants are induction wearing a programmer’s coat.

Exercises

Exercise 13.1 EASY

Prove by induction: $\sum_{i=1}^n (2i - 1) = n^2$ for $n \geq 1$.

Exercise 13.2 EASY

Prove by induction: $\sum_{k=1}^n k \cdot k! = (n + 1)! - 1$.

Exercise 13.3 EASY

Prove by induction: $\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$.

Exercise 13.4 MEDIUM

Prove that $n! > 3^n$ for all $n \geq 7$.

Exercise 13.5 MEDIUM

Prove that $6 \mid (n^3 - n)$ for all $n \geq 0$.

Exercise 13.6 MEDIUM

Prove by induction that $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

Exercise 13.7 MEDIUM

Prove Bernoulli’s inequality: $(1 + x)^n \geq 1 + nx$ for all real $x \geq -1$ and $n \in \mathbb{N}$. Where does the step use $x \geq -1$?

Exercise 13.8 MEDIUM

A chocolate bar is an $m \times n$ grid of squares. Prove (strong induction on the number of squares) that exactly $mn - 1$ breaks along grid lines are needed to reduce it to unit squares.

Exercise 13.9 HARD

Prove by strong induction that every $n \geq 1$ has a binary representation: it can be written as $\sum_i b_i 2^i$ with $b_i \in \{0, 1\}$.

Exercise 13.10 HARD

Deduce the well-ordering principle from weak induction: if $S \subseteq \mathbb{N}$ has no least element, prove by strong induction that $S = \emptyset$.

Exercise 13.11 HARD

Find the flaw: “Theorem: $a^n = 1$ for every $a \neq 0$ and $n \geq 0$. Base: $a^0 = 1$. Strong step: $a^{n+1} = \frac{a^n \cdot a^n}{a^{n-1}} = \frac{1 \cdot 1}{1} = 1$.”

Solutions to the Exercises

Solution 13.1. Base: $1 = 1^2$. Step: $\sum_{i=1}^{n+1} (2i - 1) = n^2 + (2n + 1) = (n + 1)^2$. ■

Solution 13.2. Base: $1 \cdot 1! = 1 = 2! - 1$. Step: $\sum_{k=1}^{n+1} k k! = (n + 1)! - 1 + (n + 1)(n + 1)! = (n + 1)!(n + 2) - 1 = (n + 2)! - 1$. ■

Solution 13.3. Base: $\frac{1}{2} = \frac{1}{2}$. Step: $\frac{n}{n+1} + \frac{1}{(n+1)(n+2)} = \frac{n(n+2)+1}{(n+1)(n+2)} = \frac{(n+1)^2}{(n+1)(n+2)} = \frac{n+1}{n+2}$. ■

Solution 13.4. Base: $7! = 5040 > 2187 = 3^7$. Step: for $n \geq 7$, $(n + 1)! = (n + 1)n! > (n + 1)3^n \geq 8 \cdot 3^n > 3^{n+1}$. ■

Solution 13.5. Base: $6 \mid 0$. Step: $(n + 1)^3 - (n + 1) = (n^3 - n) + 3(n^2 + n) = (n^3 - n) + 3n(n + 1)$. The first term is divisible by 6 by hypothesis; $n(n + 1)$ is even, so $3n(n + 1)$ is divisible by 6 as well. ■

Solution 13.6. Base: $1 = 1$. Step: with $T_n = \frac{n(n+1)}{2}$, $\sum_{k \leq n+1} k^3 = T_n^2 + (n + 1)^3 = \frac{(n+1)^2}{4}(n^2 + 4(n + 1)) = \frac{(n+1)^2(n+2)^2}{4} = T_{n+1}^2$. ■

Solution 13.7. Base: $(1 + x)^0 = 1 \geq 1$. Step: $(1 + x)^{n+1} = (1 + x)^n(1 + x) \geq (1 + nx)(1 + x)$ — this multiplication preserves the inequality *because* $1 + x \geq 0$, i.e. $x \geq -1$. Expanding: $1 + (n + 1)x + nx^2 \geq 1 + (n + 1)x$. ■

Solution 13.8. $P(s)$: any bar of s squares needs $s - 1$ breaks. Base $s = 1$: 0 breaks. Step: a bar of $s + 1$ squares is first broken into pieces of a and b squares, $a + b = s + 1$, $a, b \leq s$. By strong hypothesis they need $a - 1$ and $b - 1$ further breaks: total $1 + (a - 1) + (b - 1) = s = (s + 1) - 1$, independent of the chosen break. ■

Solution 13.9. Base: $1 = 2^0$. Step: let $n + 1 \geq 2$. If $n + 1$ is even, write $n + 1 = 2m$ with $1 \leq m \leq n$; by strong hypothesis $m = \sum b_i 2^i$, so $n + 1 = \sum b_i 2^{i+1}$. If odd, $n + 1 = 2m + 1$ with $0 \leq m \leq n$, and $n + 1 = 1 + \sum b_i 2^{i+1}$, the new bit $b_0 = 1$. ■

Solution 13.10. Let $S \subseteq \mathbb{N}$ have no least element; set $P(n)$: “no element of S is $\leq n$ ”. $P(0)$: if $0 \in S$ it would be least, so $0 \notin S$. Step: assume $P(n)$, i.e. $S \cap \{0, \dots, n\} = \emptyset$. If $n + 1 \in S$ it would be the least element — impossible; so $P(n + 1)$. By induction every n avoids S : $S = \emptyset$. Contrapositive: every nonempty S has a least element. ■

Solution 13.11. The step at $n = 0$ computes $a^1 = \frac{a^0 a^0}{a^{-1}}$, invoking the hypothesis at $n - 1 = -1$, which is outside the induction’s range (and the base only covers $n = 0$). The chain $P(0) \Rightarrow P(1)$ is broken, so nothing beyond the base is established. ■

Recursive Definitions

Notes By Pr. El Hadiq Zouhair

14.1 The recursive idea

A *recursive definition* specifies an object in terms of smaller instances of itself, anchored by explicit base cases. Three kinds of objects are defined this way: *functions* (a base value plus a recurrence), *sets* (base elements plus closure rules), and *structures* (atomic structures plus constructors). Induction (Chapter 13) is the proof principle that matches each kind.

14.2 Recursive functions

Definition 14.1. A function on $\mathbb{N}$ is *recursively defined* by giving $f(0)$ (and possibly further initial values) and a rule computing $f(n+1)$ from earlier values.

Example 14.2. Factorial: $0! = 1$, $(n+1)! = (n+1) \cdot n!$. Powers: $a^0 = 1$, $a^{n+1} = a \cdot a^n$. Fibonacci: $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$. Ackermann's function — recursive in two variables — grows faster than any primitive recursive function.

14.2.1 From recurrence to closed form

For *linear homogeneous* recurrences, the characteristic-equation method of Chapter 9 applies: $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ has solution $\alpha r_1^n + \beta r_2^n$ (distinct roots) or $(\alpha + \beta n)r^n$ (double root); Binet's formula $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$ is the showcase. Simple non-homogeneous patterns are handled by *unrolling* (iteration) or *telescoping*.

Example 14.3 (Telescoping). $a_1 = 1$, $a_n = a_{n-1} + (2n-1)$. Unroll: $a_n = 1 + \sum_{k=2}^n (2k-1) = \sum_{k=1}^n (2k-1) = n^2$ (Chapter 13, Exercise 13.1).

Example 14.4 (Tower of Hanoi). Moving n discs requires $H_1 = 1$, $H_n = 2H_{n-1} + 1$ moves: move $n-1$ discs aside, the largest disc, then the $n-1$ again. Unrolling, or induction, gives

$H_n = 2^n - 1$. With 64 discs: $2^{64} - 1 \approx 1.8 \times 10^{19}$ moves — the legend's monks will not finish soon.

14.3 Recursively defined sets

Definition 14.5. A set S is defined recursively by: a *basis* (specified elements of S), *recursive rules* (ways to build new elements from old), and the implicit *exclusion* clause: nothing else is in S (S is the *smallest* set satisfying basis and rules).

Example 14.6 (Even naturals). Basis: $0 \in E$. Rule: $x \in E \Rightarrow x + 2 \in E$. Then $E = \{0, 2, 4, \dots\}$, the even naturals.

Example 14.7 (Strings). The set Σ^* of strings over alphabet Σ : basis — the empty string $\lambda \in \Sigma^*$; rule — if $w \in \Sigma^*$ and $a \in \Sigma$ then $wa \in \Sigma^*$. Concatenation, length and reversal are then defined recursively on this structure, e.g. $\text{len}(\lambda) = 0$, $\text{len}(wa) = \text{len}(w) + 1$.

Example 14.8. The well-formed formulas of Chapter 2 are a recursively defined set: atoms as basis, connectives as rules — which is what makes structural induction on formulas legitimate.

14.4 Recursively defined structures

Definition 14.9. *Rooted binary trees*: basis — the empty tree; rule — a root with left and right binary subtrees forms a binary tree. *Full binary trees*: basis — a single vertex; rule — a root joined to two full binary subtrees.

14.5 Structural induction

Theorem 14.10 (Structural induction). To prove that every element of a recursively defined set has property P : prove P for each basis element, and prove that each rule preserves P (if the inputs satisfy P , so does the output).

Example 14.11 (Complete binary trees). The *complete* full binary tree of height $n \geq 1$ has $2^n - 1$ vertices, of which 2^{n-1} are leaves. Induction on n : height 1 — one vertex, one leaf: $2^1 - 1 = 1$, $2^0 = 1$. Step: the tree of height $n + 1$ is a root over two trees of height n : $2(2^n - 1) + 1 = 2^{n+1} - 1$ vertices and $2 \cdot 2^{n-1} = 2^n$ leaves.

Example 14.12 (Strings). $\text{len}(uv) = \text{len}(u) + \text{len}(v)$: structural induction on v . Base $v = \lambda$: $\text{len}(u\lambda) = \text{len}(u)$. Rule case $v = wa$: $\text{len}(u(wa)) = \text{len}((uw)a) = \text{len}(uw) + 1 = \text{len}(u) + \text{len}(w) + 1 = \text{len}(u) + \text{len}(wa)$.

14.6 Recursion in computation

14.6.1 Recursive algorithms and correctness

A recursive algorithm mirrors a recursive definition; induction proves it correct. For the recursive factorial: base — the program returns $1 = 0!$ on input 0; step — if it returns $n!$ on n , then on $n+1$ it returns $(n+1) \cdot n! = (n+1)!$. Mergesort splits a list in half, sorts each recursively and merges; strong induction on length proves correctness, and its running time obeys $T(n) = 2T(n/2) + O(n)$, solved by the *master theorem* as $T(n) = O(n \log n)$.

14.6.2 Memoization and dynamic programming

Naive recursive Fibonacci recomputes subproblems exponentially ($T(n) = T(n-1) + T(n-2) + O(1)$ is itself Fibonacci-sized). Caching each computed value — *memoization* — or filling a table bottom-up — *dynamic programming* — reduces the cost to linear. The lesson: a recursive *definition* need not dictate a naive recursive *computation*.

14.6.3 Tail recursion, mutual recursion, fixed points

A *tail-recursive* function performs its recursive call last, so it can be transformed mechanically into a loop (accumulator pattern): $\text{fact}(n, \text{acc}) = \text{fact}(n-1, n \cdot \text{acc})$. *Mutual* recursion defines functions in terms of each other, e.g. $\text{even}(0) = \text{T}$, $\text{even}(n) = \text{odd}(n-1)$; $\text{odd}(0) = \text{F}$, $\text{odd}(n) = \text{even}(n-1)$ — proved correct by simultaneous induction. More abstractly, a recursive definition presents its object as a *fixed point* of an operator ($S = F(S)$); self-similar fractals such as the Koch curve and Sierpiński triangle, and the generations of a cellular automaton (e.g. rule 90, which paints Pascal's triangle mod 2), are recursive definitions made visible.

Common recursion pitfalls: a missing or unreachable base case (infinite regress); recursive calls that do not strictly decrease toward the base; overlapping subproblems solved exponentially often when a cache would do; and confusing the definition of a set with membership testing — the exclusion clause matters.

14.7 Summary

Recursive definitions build functions (base values + recurrence), sets (basis + closure rules + exclusion) and structures (atoms + constructors). Closed forms come from characteristic equations, unrolling or telescoping; structural induction is the universal proof method, with weak/strong induction as its special case on $\mathbb{N}$. Computationally, recursion pairs with induction for correctness, with memoization for efficiency, and with iteration via tail calls.

Exercises

Exercise 14.1 EASY

Give recursive definitions of: (a) $f(n) = 5n$; (b) $g(n) = 3^n$; (c) the sequence 4, 9, 14, 19, ...

Exercise 14.2 EASY

Compute a_4 where $a_0 = 3$ and $a_n = 2a_{n-1} - 1$, then conjecture and verify a closed form.

Exercise 14.3 EASY

Define recursively the set $S = \{3, 6, 9, 12, \dots\}$ of positive multiples of 3.

Exercise 14.4 MEDIUM

Solve by unrolling: $a_1 = 2$, $a_n = a_{n-1} + 2^n$, and verify by induction.

Exercise 14.5 MEDIUM

Prove by induction that the Tower of Hanoi recurrence $H_1 = 1$, $H_n = 2H_{n-1} + 1$ has solution $H_n = 2^n - 1$.

Exercise 14.6 MEDIUM

Define the reversal of a string by $\lambda^R = \lambda$ and $(wa)^R = a w^R$. Prove by structural induction that $(uv)^R = v^R u^R$.

Exercise 14.7 MEDIUM

Let S be defined by: $5 \in S$; if $x, y \in S$ then $x + y \in S$. Determine S exactly and prove your answer by structural induction (one inclusion) and ordinary argument (the other).

Exercise 14.8 MEDIUM

A *full* binary tree is a single vertex, or a root with two full binary subtrees. Prove by structural induction that the number of leaves of a full binary tree exceeds the number of internal vertices by exactly one.

Exercise 14.9 HARD

Define $T(n) = T(\lfloor n/2 \rfloor) + 1$, $T(1) = 1$. Show $T(n) = \lfloor \log_2 n \rfloor + 1$ by strong induction.

Exercise 14.10 HARD

Naive recursive Fibonacci makes C_n function calls, where $C_0 = C_1 = 1$ and $C_n = C_{n-1} + C_{n-2} + 1$. Prove $C_n = 2F_{n+1} - 1$, confirming exponential blow-up.

Exercise 14.11 HARD

The Koch curve is defined recursively: K_0 is a segment of length 1; K_{n+1} replaces each segment of K_n by four segments each one third as long. Find the length L_n of K_n , prove it by induction, and conclude about $\lim L_n$.

Solutions to the Exercises

Solution 14.1. (a) $f(0) = 0$, $f(n+1) = f(n) + 5$. (b) $g(0) = 1$, $g(n+1) = 3g(n)$. (c) $a_0 = 4$, $a_{n+1} = a_n + 5$. ■

Solution 14.2. $a_1 = 5$, $a_2 = 9$, $a_3 = 17$, $a_4 = 33$. Conjecture $a_n = 2^{n+1} + 1$: base $a_0 = 3$; step $2(2^{n+1} + 1) - 1 = 2^{n+2} + 1$. ✓ ■

Solution 14.3. Basis: $3 \in S$. Rule: $x \in S \Rightarrow x + 3 \in S$. (Exclusion clause: nothing else.) An easy induction shows $S = \{3k \mid k \geq 1\}$. ■

Solution 14.4. $a_n = 2 + \sum_{k=2}^n 2^k = 2 + (2^{n+1} - 4) = 2^{n+1} - 2$. Induction: base $a_1 = 2 = 2^2 - 2$; step $a_{n+1} = (2^{n+1} - 2) + 2^{n+1} = 2^{n+2} - 2$. ■

Solution 14.5. Base: $H_1 = 1 = 2^1 - 1$. Step: $H_{n+1} = 2H_n + 1 = 2(2^n - 1) + 1 = 2^{n+1} - 1$. ■

Solution 14.6. Structural induction on v . Base $v = \lambda$: $(u\lambda)^R = u^R = \lambda^R u^R$. Step $v = wa$: $(u(wa))^R = ((uw)a)^R = a(uw)^R = a(w^R u^R) = (aw^R)u^R = (wa)^R u^R$. ■

Solution 14.7. $S = \{5k \mid k \geq 1\}$, the positive multiples of 5. ($\subseteq$) Structural induction: the basis element 5 is a multiple of 5, and the sum of two multiples of 5 is one. ($\supseteq$) Induction on k : $5 \in S$; if $5k \in S$ then $5(k+1) = 5k + 5 \in S$ by the rule. ■

Solution 14.8. $P(T)$: $\text{leaves}(T) = \text{internal}(T) + 1$. Base: a single vertex has 1 leaf, 0 internal: ✓. Step: $T = \text{root over } T_1, T_2 \text{ satisfying } P$. Then $\text{leaves}(T) = \text{leaves}(T_1) + \text{leaves}(T_2) = \text{internal}(T_1) + \text{internal}(T_2) + 2 = \text{internal}(T) + 1$, since $\text{internal}(T) = \text{internal}(T_1) + \text{internal}(T_2) + 1$ (the new root). ■

Solution 14.9. $P(n)$: $T(n) = \lfloor \log_2 n \rfloor + 1$. Base: $T(1) = 1 = 0 + 1$. Step: let $n \geq 2$, so $1 \leq \lfloor n/2 \rfloor < n$; by strong hypothesis $T(n) = \lfloor \log_2 \lfloor n/2 \rfloor \rfloor + 2$. Writing $2^k \leq n < 2^{k+1}$ (so $\lfloor \log_2 n \rfloor = k$), we get $2^{k-1} \leq \lfloor n/2 \rfloor < 2^k$, whence $\lfloor \log_2 \lfloor n/2 \rfloor \rfloor = k - 1$ and $T(n) = (k - 1) + 2 = k + 1$. ✓ ■

Solution 14.10. Induction with two bases: $C_0 = 1 = 2F_1 - 1$, $C_1 = 1 = 2F_2 - 1$. Step: $C_n = C_{n-1} + C_{n-2} + 1 = (2F_n - 1) + (2F_{n-1} - 1) + 1 = 2F_{n+1} - 1$. Since $F_{n+1} \sim \varphi^{n+1}/\sqrt{5}$, the call count grows exponentially in n . ■

Solution 14.11. Each step multiplies the number of segments by 4 and divides their length by 3, so $L_n = (4/3)^n$. Induction: $L_0 = 1$; step $L_{n+1} = 4 \cdot \frac{L_n}{3} \cdot \frac{(\text{per segment})}{(\text{same})} = \frac{4}{3}L_n = (4/3)^{n+1}$. As $4/3 > 1$, $L_n \rightarrow \infty$: the limiting curve has infinite length (while spanning a bounded region — the hallmark of a fractal). ■

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